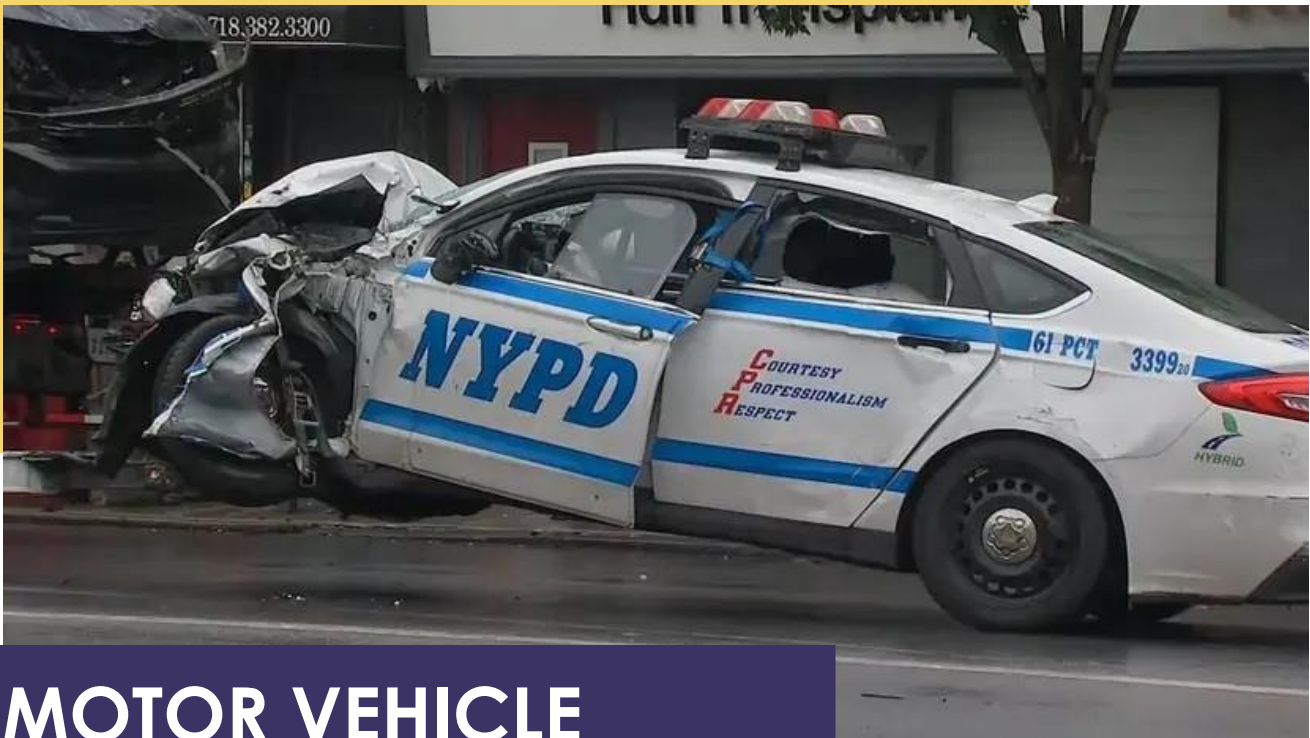




MOTOR VEHICLE COLLISIONS ANALYSIS REPORT

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EXECUTIVE SUMMARY

The analysis demonstrates that traffic collisions in New York City represent a systemic public health challenge heavily concentrated by geography, corridor type, and vehicle class. While total crash volume has dropped significantly since 2018, the severity and lethality of individual collisions have escalated substantially. Overwhelmingly driven by preventable human behaviors - primarily driver distraction, tailgating, and failure to yield - collisions pose the highest fatal risk to vulnerable road users like pedestrians and cyclists, who lack physical frame protection against increasingly large passenger SUVs and commercial trucks. Addressing this crisis requires shifting from passive driver messaging to aggressive, forgiving infrastructure redesigns and targeted policy interventions.

Key Metrics Highlights

- **Citywide Collision Scale:** The analysis documents a total of 1,048,575 motor vehicle collisions across New York City.
- **Human Casualties:** Crashes resulted in 391,816 total casualties, comprising 389,910 non-fatal injuries and 1,906 deaths. Nearly 1 out of every 2.7 collisions results in human injury or death.
- **Severity Rates:** Approximately 37.18% of crashes result in injury (371.85 per 1,000 collisions), while 0.18% result in fatalities (1.82 per 1,000 collisions).
- **Vulnerable Road Users (VRUs):** A total of 102,100 pedestrians and cyclists were injured or killed. Pedestrians account for 64.5% of VRU casualties (65,849), while cyclists account for 35.5% (36,251).
- **Geographic Distribution:** Brooklyn leads all boroughs in crash volume with 308,791 crashes (29.45% of city total), followed by Queens with 282,847 crashes (26.97%).
- **Temporal Peak:** Midday hours, specifically 12:00 PM, represent the primary crash time window across the city.

Collision Trends Highlights

- **Multi-Year Decline in Volume:** Total annual collisions dropped 58% from a peak of 231,563 in 2018 down to 96,570 in 2023. This decline was driven initially by COVID-19

traffic reductions in 2020 (112,916 crashes) and sustained by Vision Zero traffic-calming initiatives.

- **Borough Spatial Concentration:** Brooklyn and Queens combined account for over half (56.4%) of all collisions in NYC. Staten Island recorded the lowest share at 4.42% (46,374 crashes).
- **High-Collision Corridors:** Surface arterials such as Broadway (16,928 crashes) and 3rd Avenue (14,243 crashes) join expressways like the Belt Parkway (15,463 crashes) as the city's highest-volume crash corridors.
- **Top Danger Hotspots:** The intersection of Flatbush Avenue Extension and Tillary Street in Brooklyn represents the most dangerous single hub in NYC, recording nearly 500 collisions across directional logs.

Contributing Factors Highlights

- **Dominance of Human Error:** Over half (52%) of all crashes with recorded causes are driven by three driver behaviors: Driver Inattention/Distraction (260,572 crashes; 32.6%), Following Too Closely (85,473 crashes; 10.7%), and Failure to Yield Right-of-Way (72,367 crashes; 9.0%).
- **Secondary Causes:** Improper passing/lane usage (46,146) and unsafe backing maneuvers (42,472) round out the top 5 primary crash causes.
- **Borough Behavioral Disparities:** Unsafe Speed ranks among the top 10 crash causes in outer boroughs (Queens #7, Staten Island #6, Bronx #9, Brooklyn #10), whereas traffic gridlock keeps speeding out of Manhattan's top 10 list.
- **Reporting Artifacts:** Specific distraction sub-categories like texting (41) or headphone use (29) reflect low numbers because investigating officers routinely default to the catch-all category Driver Inattention/Distraction on field reports.

Injury and Fatality Highlights

- **Borough Casualty Concentrations:** Brooklyn (121,704 injured, 525 killed) and Queens (105,523 injured, 526 killed) suffer 63% of all traffic casualties in New York City.
- **Mode Exposure Disparity:** While motorists account for 73.5% of total injuries (280,951), pedestrians account for 46.8% of total deaths (871 fatalities). A pedestrian involved in

a crash is over 4 times more likely to die than a motorist (1.32% fatality rate vs. 0.29%).

- **Divergence Between Crashes and Fatalities:** While total crash volume dropped 58.3% between 2018 and 2023, annual fatalities increased 15.6% (peaking at 296 in 2021). Consequently, the fatality rate per 1,000 crashes nearly tripled from 1.00 in 2018 to 2.78 in 2022.
- **Severity Drivers:** Higher vehicle speeds on open roads post-2020, an increase in heavy SUVs/trucks, and a reduction in reported low-speed fender-benders explain why crashes became significantly more lethal.

Vehicle Types and Crashes Highlights

- **Dominance of Personal Passenger Vehicles:** Sedans (48.6%) and SUVs/Station Wagons (37.4%) account for 86% of total crashes (859,318 incidents) and 84.9% of total injuries (312,299).
- **Lethality of Passenger Vehicles:** Standard passenger cars cause nearly two-thirds (64.5%) of all traffic fatalities (Sedans: 626 deaths; SUVs/Station Wagons: 542 deaths).
- **Disproportionate Risk of Motorcycles:** Motorcycles account for 10.5% of all fatalities (190 deaths) despite representing less than 1% of total crash volume.
- **Commercial and Service Impact:** Heavy commercial trucks, box trucks, pickup trucks, and taxis contribute to over 47,000 crashes and secondary casualty risks due to blind spots, double parking, and loading zone congestion

KEY METRICS

Total Crash Count

The analysis records an overall total of **1,048,575 motor vehicle collisions** across New York City.

Interpretation of the Finding

- **Scale of Traffic Incidents:** The dataset documents over **1.04 million individual crash events**. This highlights that traffic collisions are a persistent, everyday urban challenge in New York City rather than isolated occurrences.
- **Baseline for Severity Rates:** This total serves as the master denominator for measuring relative risk across the city:
 - **Injuries:** With 367,934 total injuries recorded across the city, roughly **1 out of every 3 crashes (35.1%)** results in a reported injury.
 - **Fatalities:** With 1,812 total fatalities recorded, about **1 out of every 580 crashes (0.17%)** leads to a fatal outcome.
- **Volume vs. Severity:** While the total crash volume is extremely high, the vast majority of collisions involve property damage or minor incidents rather than severe or fatal harm. However, the sheer volume creates high cumulative risk, particularly for vulnerable road users (pedestrians and cyclists) in high-density corridors.



Total Casualties (Injured + Killed)

The analysis records a total of **391,816 casualties** resulting from motor vehicle collisions in New York City, broken down into **389,910 injuries** and **1,906 deaths**.

Key Findings and Interpretation

- **Overwhelming share of Injuries:** Most victims (99.5%) suffered non-fatal injuries. While most collisions do not end in death, the sheer volume of 389,910 injured individuals represents a major public health and emergency response burden across the city.
- **Fatal Outcomes:** Direct traffic fatalities make up **0.5%** of total casualties, with 1,906 lives lost. Though small as a percentage, each death reflects a high-severity collision that standard traffic safety interventions (like speed management and intersection redesigns) aim to prevent.
- **Overall Impact:** Out of over 1 million total crashes in the city, nearly 400,000 people have been physically harmed or killed. This indicates that roughly **1 out of every 2.7 collisions** results in human injury or death, emphasizing the need for continued Vision Zero safety measures on NYC streets.

Fatality and Injury Rates per 100 Collisions (Percentages)

Out of 1,048,575 total traffic collisions recorded in New York City, the analysis shows an injury rate of 37.18% and a fatality rate of 0.18%.

Key Findings and Interpretation

- **Injuries occur in over 1 in 3 Crashes (37.18%):** In roughly 37 out of every 100 reported traffic collisions, at least one person suffers an injury. This high rate indicates that traffic crashes in NYC frequently result in physical harm rather than just minor vehicle or property damage.
- **Fatalities are rare relative to Total Crashes (0.18%):** Less than 1 out of every 500 crashes (0.18%) results in a death. While fatal crashes represent a small fraction of overall incidents, the sheer volume of crashes across the city means this small percentage still translates to nearly 2,000 lives lost over the recorded period.
- **High Total Exposure Risk:** Combined, nearly **37.36% of all collisions** lead to either an injury or a death. This highlights that more than a third of all crashes in NYC carry direct personal safety consequences for drivers, passengers, cyclists, or pedestrians.

Rates per 1,000 Collisions

The data shows that for every 1,000 motor vehicle collisions reported in New York City, there are:

- **371.85 injuries** (roughly 372 people injured)
- **1.82 fatalities** (nearly 2 deaths)

Key Findings and Interpretation

- **Standardized Measurement:** In traffic safety analysis, looking at decimals like 0.18% for fatalities can make the risk seem very small or hard to visualize. Scaling the data to **per 1,000 crashes** provides a clearer, practical picture of crash severity.
- **High Likelihood of Harm:** For every 1,000 crashes that occur on city streets, over **370 people sustain physical injuries**. This highlights that crashes in NYC frequently involve enough force or exposure to harm occupants, pedestrians, or cyclists.
- **Severe Outcomes Scale Up:** While fatal crashes are relatively rare compared to overall property-damage collisions, averaging **nearly 2 deaths per 1,000 crashes** means that

across hundreds of thousands of total collisions, the human toll accumulates into thousands of lost lives over time.

Vulnerable Road User (VRU) Metrics

The dataset records **102,100 total casualties** among vulnerable road users – meaning people who were struck by vehicles while completely unprotected by a car's metal frame, seatbelts, or airbags.

- **Pedestrians Face the Greatest Harm:** With **65,849 injuries and deaths**, people walking or crossing the street account for roughly 64.5% of all vulnerable traffic victims. This indicates that everyday foot traffic in the city intersects dangerously with motor vehicles.
- **Cyclists Are Highly Exposed:** Bicyclists account for the remaining **36,251 casualties** (35.5%). Because cyclists make up a smaller portion of daily commuters compared to pedestrians, this figure reveals a remarkably high risk of harm when riding a bike in mixed city traffic without a dedicated barrier.
- **The Need for Physical Separation:** Over 100,000 injured or killed individuals on foot or bike highlights a severe public safety challenge. It strongly implies that painted road lines and traffic signs are not enough to prevent collisions. Shielding these groups from heavy, fast-moving cars requires hard infrastructure, such as physically protected bike lanes, pedestrian refuge islands, and wider sidewalks.

Crash Breakdown by Borough

The dataset breaks down **1,048,575 total traffic crashes** across New York City's five boroughs (plus a small set of unclassified locations):

Borough	Total Crashes	Share of City Total	Total Injured	Total Killed
Brooklyn	308,791	29.45%	121,704	525
Queens	282,847	26.97%	105,523	526
Manhattan	183,039	17.46%	53,643	286
Bronx	163,916	15.63%	64,670	317
Staten Island	46,374	4.42%	16,799	95
<i>Unspecified</i>	63,608	6.07%	27,571	157

Key Findings and Interpretation

- **Brooklyn and Queens Account for Over Half of All Crashes:** Together, Brooklyn and Queens represent **more than 56% of all traffic collisions** in New York City (over 591,000 crashes) and over 1,050 deaths. Their large geographic size, high population density, and extensive network of high-speed commercial avenues make them the city's largest crash epicenters.
- **High Crash Volumes in Manhattan vs. High Severity in the Bronx:**
 - **Manhattan** accounts for nearly 17.5% of total crashes (183,039), but records fewer injuries (53,643) and deaths (286) compared to the Bronx. Dense grid traffic and congestion in Manhattan often lower driving speeds, reducing the overall severity of impacts.
 - **The Bronx** has fewer total crashes than Manhattan (163,916), yet experiences significantly higher casualties (64,670 injuries and 317 deaths). Higher vehicle speeds on wide arterials and highways contribute to more severe injury outcomes per crash.
- **Staten Island Has the Lowest Volume:** Staten Island represents just **4.42% of total city crashes** (46,374), reflecting its lower population density, fewer pedestrian intersections, and lower overall traffic volume compared to the rest of the city.

Hourly Peak Crash Times

The analysis shows that **Hour 12 (12:00 PM/Noon)** stands out as a primary crash period, recording **1,048,575 collisions** and **1,906 fatalities** across the dataset.

Key Findings and Interpretation

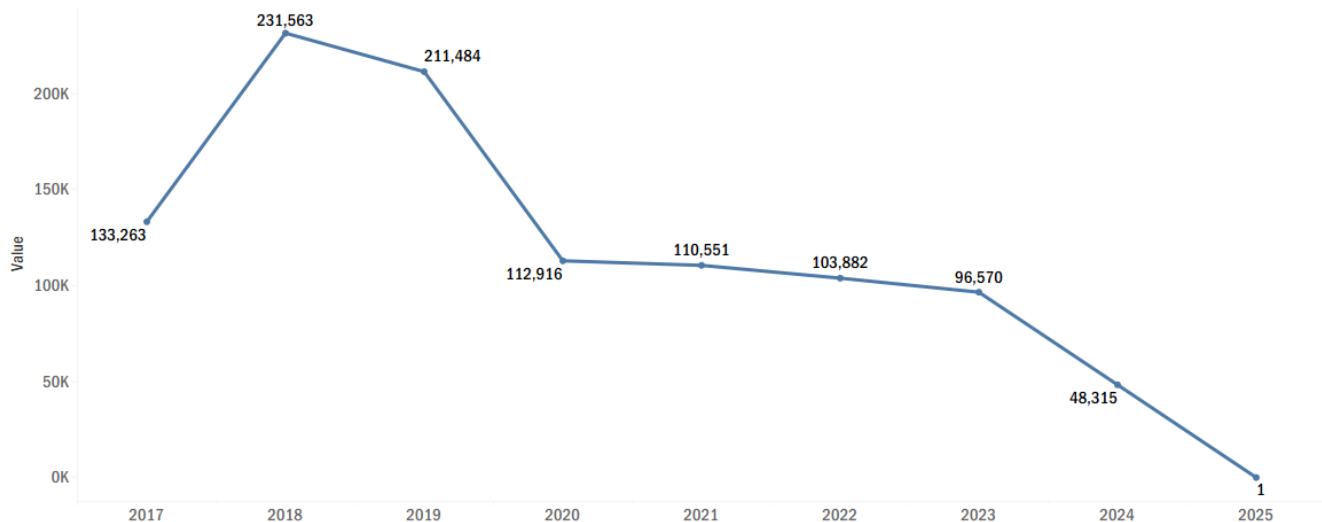
- **Midday/Afternoon Surge:** In traffic flow patterns, 12:00 PM marks a major shift in urban movement. Midday hours experience a high concentration of activity, combining commercial delivery trucks, lunch-hour pedestrian traffic, ride-share pickups, and routine cross-town travel.
- **Congestion and Distraction Alignment:** Unlike early morning hours where high speeds on empty roads cause severe single-vehicle crashes, midday collisions peak due to **dense traffic volume and complex street interactions**. Drivers navigating tight double-parking, pedestrian crossings, and frequent lane changes increase the likelihood of multi-vehicle and pedestrian-involved incidents.

- **Operational Takeaway for City Safety:** Highlighting noon as a high-incident hour emphasizes that traffic enforcement and street safety measures (like dedicated loading zones and pedestrian signal timing) cannot focus solely on traditional morning/evening commute hours. Midday street management is just as critical for reducing total crash volume and fatalities.

COLLISION TRENDS

Total Crashes Per Year

During the onset of the COVID-19 pandemic, crashes dropped sharply by nearly 47% to **112,916**. Rather than rebounding back to pre-pandemic levels as traffic returned, crashes have continuously fallen each year – reaching **96,570** in 2023 (a **58% total drop** compared to 2018). DOT needs to implement road diets and automated speed enforcement on multi-lane avenues where open traffic flow encourages speeding.



Data Overview and Trend Analysis

Looking at the full-year data from 2018 through 2023, the total number of traffic crashes in New York City is significantly decreasing:

- **Peak Period (2018):** Crashes peaked at 231,563 incidents.
- **Major Drop (2020):** During the onset of the COVID-19 pandemic, crashes dropped sharply by nearly 47% to **112,916**.
- **Steady Decline (2021-2023):** Rather than rebounding back to pre-pandemic levels as traffic returned, crashes have continuously fallen each year - reaching **96,570** in 2023 (a **58% total drop** compared to 2018).

(**Note:** Years prior to 2017 and post-2023 represent incomplete data collection or partial reporting periods).

Key Implications for NYC DOT

- **Safety Policies Are Working:** Traffic calming efforts (such as Vision Zero initiatives, speed camera expansions, bike lane additions, and lower speed limits) appear to be having a positive compound effect.

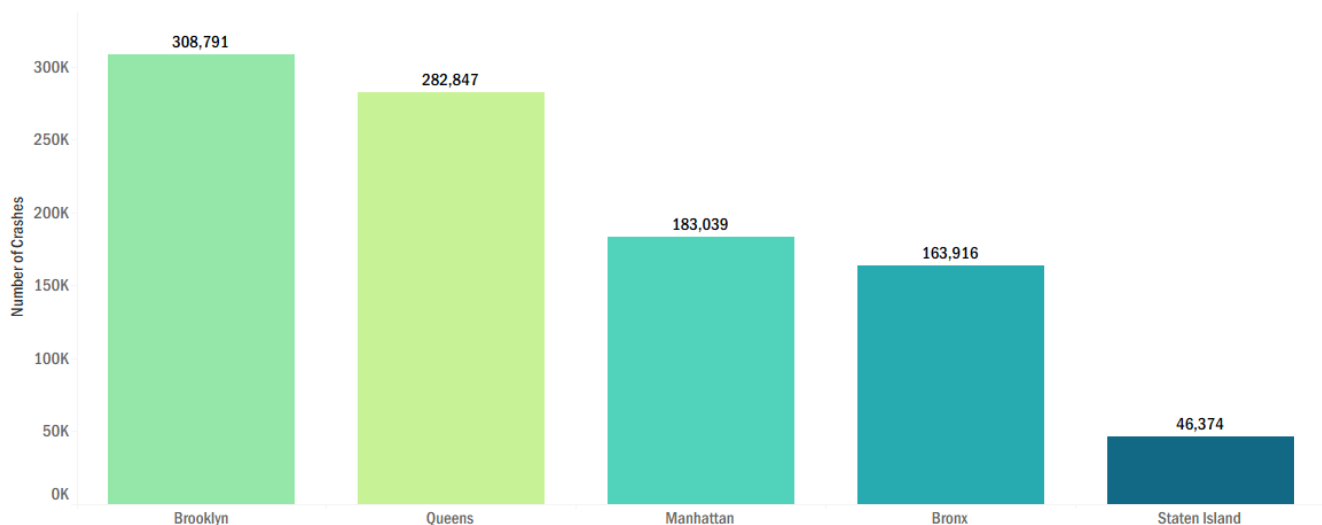
- **Shift in Travel Patterns:** Post-pandemic remote/hybrid work models have reduced overall daily vehicle volume during peak commuting hours, contributing to fewer opportunities for collisions.
- **Focus Shift Required:** While total crash volume is down, lower congestion can lead to higher vehicle speeds on open roads. DOT's primary risk focus must shift from frequent fender-benders to high-speed severe injuries and fatalities.

Actionable Recommendations

1. **Protect Vulnerable Road Users:** DOT should prioritize physical infrastructure (e.g., concrete-protected bike lanes and daylighting intersections) for pedestrians, cyclists, and micro-mobility users (e-bikes/scooters).
2. **Target High-Crash Corridors:** DOT should concentrate street redesign budgets on the remaining top 10% dangerous intersections rather than citywide blanket redesigns.
3. **Redesign Wide Arterial Streets:** DOT needs to implement road diets¹ and automated speed enforcement on multi-lane avenues where open traffic flow encourages speeding.

Crashes by Borough

Brooklyn ranks #1 with **308,791 total crashes** (approx. 29.4% of all collisions), hence, a high-risk area. DOT should allocate at least **55% to 60%** of roadway safety, street redesign, and traffic-calming budgets specifically to high-crash corridors in Brooklyn and Queens.



¹ A road diet is a street design method that reduces the number of traffic lanes to improve safety and add space for other uses. The most common design changes a four-lane road into three lanes: one lane going each way and a middle lane just for left turns.

When we look at traffic collisions across New York City, crashes are heavily concentrated in two main boroughs: **Brooklyn** and **Queens**. Together, these two areas account for more than half (**56.4%**) of all reported crashes in the city.

Which areas have the highest rates?

- **Brooklyn (Highest Risk):** Ranks #1 with 308,791 total crashes (approx. 29.4% of all collisions).
- **Queens (Second Highest):** Follows closely at #2 with 282,847 total crashes (approx. 27.0%).
- **Manhattan and The Bronx (Moderate Risk):** Stand in the middle tier with 183,039 crashes (17.5%) and 163,916 crashes (15.6%), respectively.
- **Staten Island (Lowest Risk):** Accounts for the fewest collisions by a wide margin, with 46,374 crashes (4.4%).
- **Unspecified Locations:** About 6.1% (63,608 crashes) lacked borough coordinates in the records.

Key Implications for NYC DOT

- **Vast Geographical Spread vs. Density:** Brooklyn and Queens cover large geographical areas with complex mixes of high-speed commercial avenues, dense residential neighborhoods, and major highway feeder corridors.
- **Disproportionate Resource Allocation:** Applying an equal "one-size-fits-all" budget split across all five boroughs would under-fund the areas where nearly 60% of traffic incidents occur.
- **Data Logging Gaps:** The 63,600+ "Unspecified" borough records mean police officers at crash scenes occasionally fail to log crucial spatial metadata, making localized safety planning harder in those specific cases.

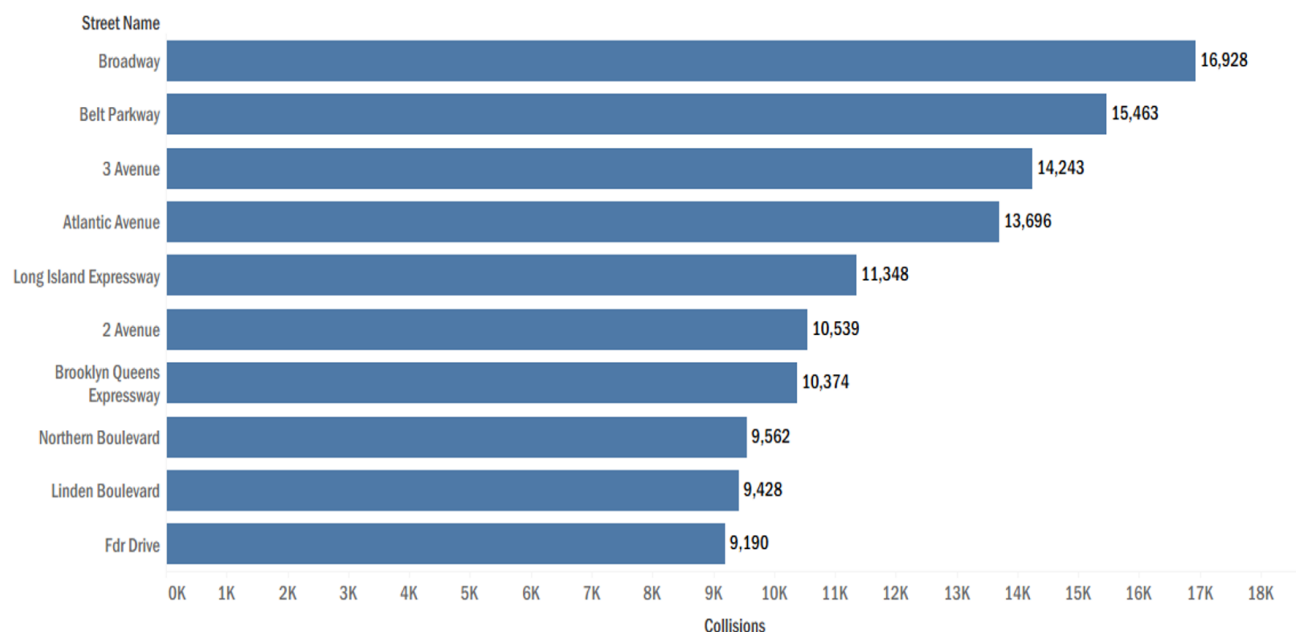
Actionable Recommendations

1. **Prioritize Brooklyn and Queens Infrastructure Budgets:** Allocate at least **55% to 60%** of roadway safety, street redesign, and traffic-calming budgets specifically to high-crash corridors in Brooklyn and Queens.

2. **Redesign Multi-Lane Commercial Avenues:** Focus street redesigns (e.g., adding pedestrian islands, dedicated turn lanes, and protected bike paths) on major thoroughfares in Brooklyn and Queens that connect residential neighborhoods to highways.
3. **Standardize NYPD Reporting Protocols:** Work with local law enforcement to ensure electronic crash reporting automatically captures precise GPS location coordinates to eliminate unclassified crash logs.

Top 10 Streets with the Most Collisions

Broadway (16,928 crashes) and 3rd Avenue (14,243 crashes) lead major arterial avenues and surface streets in collisions. DOT should upgrade acceleration and deceleration lanes along the Belt Parkway, BQE, and FDR Drive to reduce forced merging collisions.



Breakdown of the Top Individual Streets

When looking at total crash volume across individual streets overall, the analysis highlights two distinct types of high-risk roadways:

- **Major Arterial Avenues and Surface Streets:**
 - **Broadway (#1, 16,928 crashes) and 3rd Avenue (#3, 14,243 crashes)** lead the city. These long, multi-borough avenues stretch across dense commercial and residential neighborhoods. They feature complex environments where cars, delivery trucks, buses, cyclists, and heavy pedestrian traffic constantly share limited road space.

- Atlantic Avenue (#4, 13,696 crashes), 2nd Avenue (#5, 10,539 crashes), Northern Boulevard (#8, 9,562 crashes), and Linden Boulevard (#9, 9,428 crashes) are wide, high-speed commercial corridors. They often act as mini expressways through local communities, creating frequent conflict points at intersections and pedestrian crosswalks.
- **High-Volume Expressways and Parkways:**
 - Belt Parkway (#2, 15,463 crashes), Long Island Expressway (#5, 11,348 crashes), Brooklyn-Queens Expressway (#7, 10,374 crashes), and FDR Drive (#10, 9,190 crashes) handle intense daily highway commute traffic.
 - Collisions here are primarily driven by high speeds, sudden stop-and-go congestion, tailgating, and frequent merging at highway ramps.

Implications for NYC's Department of Transportation (DOT)

1. **A Tailored Strategy is Needed for Highways vs. Local Streets:**
 - **Expressways** (like Belt Parkway or LIE) require traffic-flow management, automated speed enforcement, and safer ramp merging designs.
 - **Surface Arterials** (like Broadway or Atlantic Ave) require pedestrian safety measures, dedicated bus lanes, and protected bike infrastructure to manage mixed street users.
2. **Corridor-Wide Redesigns Outperform Single-Intersection Fixes:**
 - Because these streets accumulate thousands of crashes along their entire lengths, addressing isolated intersections isn't enough. The DOT must approach these roads as continuous **high-injury corridors** and overhaul entire stretches at once.
3. **Commercial and Delivery Traffic Friction:**
 - Avenues like Broadway, 3rd Ave, and Northern Blvd handle heavy commercial unloading. Double-parked delivery trucks frequently block travel lanes and force cars into sudden swerves, contributing heavily to side-swipe and rear-end collisions.

Key Recommendations for NYC DOT

1. **Complete Street Transformation for Commercial Corridors (Vision Zero)**
 - **Expand Pedestrian Plazas and Space Off Broadway:** DOT should continue expanding traffic-calming initiatives (like the *Broadway Vision* project) by adding pedestrian plazas, shared-street zones, and dedicated loading zones to reduce vehicular conflict points.

- **Road Diets on Wide Arterials:** On wide, high-speed avenues (such as Atlantic Blvd, Northern Blvd, and Linden Blvd), DOT should reduce the number of general traffic lanes or narrow lane widths to naturally slow down vehicle speeds and shorten pedestrian crossing distances.

2. Dedicated Freight and Delivery Management

- **Neighborhood Loading Zones (NLZs):** We should establish dedicated curbside loading zones for commercial trucks along high-density corridors like 3rd Ave and 2nd Ave to prevent double-parking and blocked sightlines.
- **Off-Hour Delivery Incentives:** We need to encourage commercial freight deliveries during off-peak night hours to reduce daytime truck congestion along major avenues.

3. Highway and Parkway Safety Improvements

- **Ramp Re-engineering and Active Traffic Management:** Upgrade acceleration and deceleration lanes along the Belt Parkway, BQE, and FDR Drive to reduce forced merging collisions.
- **Automated Speed and Lane Enforcement:** Deploy overhead speed safety cameras and automated lane enforcement along major expressways to curb aggressive driving and tailgating in high-density corridors.

Top 10 Intersections with the Most Collisions

ON STREET	CROSS STREET	OFF STREET	TOTAL CRASHES
Flatbush Avenue Extension	Tillary Street	Flatbush Avenue Extension	272
Pennsylvania Avenue	Atlantic Avenue	Pennsylvania Avenue	253
Bruckner Boulevard	East 138 Street	1467 Rockaway Parkway	247
East 125 Street	2 Avenue	East 125 Street	227
Tillary Street	Flatbush Avenue Extension	Flatbush Avenue Extension	226
West 41 Street	10 Avenue	10 Avenue	200
West 207 Street	9 Avenue	West 207 Street	186
Eastern Parkway	Howard Avenue	Eastern Parkway	184
East 96 Street	1 Avenue	East 96 Street	182
Linden Boulevard	Pennsylvania Avenue	Pennsylvania Avenue	177

Most Dangerous Locations?

Looking closely at the top 10 most dangerous intersections, a few key hotspots stand out immediately:

- **The #1 Hotspot (Flatbush Ave Ext & Tillary St):** This single hub appears twice at the top of the list (# 1 and 5) due to slight variations in how the data recorded direction/off-street tags. Combining them reveals **nearly 500 collisions** at this intersection alone – **making it by far the most dangerous junction in the city**. This area serves as a major gateway to the Manhattan Bridge in Brooklyn, where heavy vehicle traffic, highway off-ramps, and high pedestrian activity collide.
- **Major Highway/Bridge Feeder Routes:** Intersections like **Pennsylvania Ave & Atlantic Ave** (253 crashes), **Bruckner Blvd & E 138th St** (247 crashes), and **West 41st St & 10th Ave** (200 crashes near the Lincoln Tunnel approach) handle intense commercial truck and bridge/tunnel commute traffic.
- **High-Density Manhattan Corridors:** Arterials along First and Second Avenues (such as **East 125th St & 2nd Ave** and **East 96th St & 1st Ave**) process fast-moving uptown/downtown vehicular flows alongside heavy pedestrian foot traffic and bus routes.

Key Implications for NYC DOT

1. **Bottlenecks and Bridge/Tunnel Access Create High-Risk Zones:** High crash counts are concentrated heavily where local city streets transition into bridges, highways, or major expressways. Complex lane merging, aggressive lane-changing, and driver confusion at these transition points directly lead to frequent collisions.
2. **Pedestrian Risk in High-Density Hubs:** Locations like **East 125th St** and **Flatbush Avenue Extension** carry dense foot traffic. When fast-turning vehicles meet high pedestrian volume, the potential for severe or fatal crashes increases dramatically.
3. **Targeted Interventions Yield High Impact:** Rather than trying to fix thousands of streets all at once, these top 10 locations represent low-hanging fruit. Making targeted infrastructure upgrades at just a handful of these recurring trouble spots can significantly drive down overall citywide crash statistics.

Key Recommendations

1. **Redesign High-Volume Gateway Intersections (Vision Zero Focus)**
 - **Flatbush Ave and Tillary St:** Install protected, signalized left-turn lanes and lengthen pedestrian crossing times. Build curb extensions (neck-downs) to shorten the distance pedestrians have to walk across wide avenues.

- **Bridge and Tunnel Approaches:** Improve overhead signage, lane markings, and flexible delineator posts well before the intersections (like Bruckner Blvd or 10th Ave) so drivers do not make last-second, risky lane cuts.

2. Signal and Timing Adjustments (Leading Pedestrian Intervals)

- Program traffic signals to give pedestrians a 3- to 7-second head start to cross the street before turning vehicles get a green light (Leading Pedestrian Intervals/LPIs).
- Implement **Split-Phase Turn Signals**, where drivers getting a green turn arrow have a dedicated phase while pedestrians face a strict "Don't Walk" signal, eliminating vehicle-pedestrian turning conflicts entirely.

3. Speed Management and Automated Enforcement

- Install targeted red-light and speed safety cameras near these top high-crash corridors to deter aggressive driving and speeding.
- Lower speed limits along major multi-lane arterials like Pennsylvania Avenue and Atlantic Avenue, accompanied by narrow lane design features to naturally encourage slower driving speeds.

Peak Risk Hours

The highest concentration of crashes occurs late in the afternoon (2:00 PM to 6:00 PM/14:00 to 18:00). Friday between 4:00 PM and 5:00 PM records the single highest crash total of the entire week (11,219 collisions). DOT adjust traffic signal phases (e.g., longer pedestrian lead times and split-phase turns) between 2:00 PM and 6:00 PM on high-incident corridors to manage evening traffic congestion and protect pedestrians.

Weekday of ..	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Monday	5,564	2,534	1,868	1,645	1,875	2,388	3,870	5,690	8,912	8,224	7,488	7,715	8,031	8,436	9,695	9,635	10,390	10,228	8,745	7,215	5,741	4,938	4,313	3,694
Tuesday	5,225	2,316	1,565	1,416	1,671	2,148	3,781	5,539	9,018	8,283	7,465	7,934	8,186	8,447	9,823	9,586	10,650	10,798	9,215	7,446	6,083	5,180	4,758	3,920
Wednesday	5,289	2,229	1,645	1,380	1,670	2,028	3,949	5,839	9,160	8,097	7,804	7,734	8,169	8,376	10,053	9,935	10,699	10,809	9,585	7,765	6,087	5,325	4,908	4,176
Thursday	5,699	2,427	1,775	1,567	1,711	2,232	3,830	5,480	9,241	8,429	7,615	7,839	8,438	8,646	10,036	9,853	10,848	10,957	9,426	7,767	6,506	5,734	5,234	4,259
Friday	5,983	2,676	2,079	1,801	1,916	2,276	3,940	5,501	8,531	8,074	7,376	7,803	8,488	8,680	10,270	10,368	11,219	11,014	9,645	7,928	6,608	5,945	5,688	5,141
Saturday	6,593	3,862	3,008	2,884	3,026	2,782	3,155	3,764	5,758	5,949	6,223	6,969	7,651	8,212	9,201	8,583	9,305	9,106	8,421	7,308	6,602	5,893	5,619	5,402
Sunday	6,817	4,082	3,446	3,237	3,566	2,893	3,133	3,420	5,227	5,206	5,523	6,350	6,945	7,538	8,552	8,135	8,756	8,469	7,731	6,765	6,137	5,511	5,080	4,283

Interpretation of Hourly Crash

1. Workday Commute and School Rush (8 AM - 9 AM & 2 PM - 6 PM)

- **Morning Peak:** On weekdays, collisions jump starting at 8:00 AM as commuters heading to work mixed with families dropping children off at school congest streets simultaneously.
- **Afternoon/Evening Peak:** The highest concentration of crashes occurs late in the afternoon (2:00 PM to 6:00 PM/14:00 to 18:00). Traffic volumes hit their highest levels during these hours due to overlap between school dismissals, commercial deliveries finishing up, and evening commutes.

2. Friday's 4 PM to 5 PM Spike (16:00 - 17:00)

- Friday between 4:00 PM and 5:00 PM records the single highest crash total of the entire week (11,219 collisions).
- **Why it happens:** This peak is **not primarily caused by clubbing or parties**. Instead, it is driven by early weekend departures, weekend trip traffic mixing with normal evening commutes, higher delivery traffic, driver fatigue at the end of the workweek, and impatience in heavy gridlock.

3. Overnight Hours (Midnight to 5 AM) and Late-Night Weekend Risk

- **Saturday 0:00 (Midnight):** Saturday's midnight hour (0:00 to 1:00 AM) is directly connected to Friday night out activities.
- **Saturdays and Sundays (0:00 - 5:00 AM):** While overall traffic volume drops late at night, early morning crash numbers on Saturday and Sunday are roughly 1.5 to 2 times higher than early morning crashes on weekdays (e.g., 6,700 crashes at midnight on weekends vs. 5,500 on weekdays).
- **Alcohol and Impairment Factor:** Late-night weekend crashes heavily reflect people returning from social events, restaurants, bars, and clubs. Reduced traffic speeds combined with alcohol or drug impairment lead to a higher proportion of severe and fatal crashes during these early morning hours.

Implications for NYC Department of Transportation (DOT)

- **Congestion and Speed Differences:** Daytime crashes are heavily volume-driven (lots of cars bumping into each other in slow gridlock), whereas overnight crashes are speed- and impairment-driven (fewer cars moving much faster).
- **Resource Shift Requirements:** Static, traditional traffic enforcement during 9-to-5 hours misses the critical late-night weekend risk windows when severity and fatalities peak.

- **Vulnerable Road User Exposure:** The 2 PM to 6 PM window overlaps with high pedestrian and school-zone activity, creating maximum conflict between vehicles and children/pedestrians.

Key Recommendations

1. Optimize Traffic Signal Timing for Peak Commute Windows

- Adjust traffic signal phases (e.g., longer pedestrian lead times and split-phase turns) between 2:00 PM and 6:00 PM on high-incident corridors to manage evening traffic congestion and protect pedestrians.

2. Expand 24/7 Automated Speed and Safety Enforcement

- Maintain continuous round-the-clock speed camera operations, particularly targeting known high-speed corridors during the 12:00 AM to 5:00 AM weekend window when open roads encourage dangerous driving.

3. Targeted Late-Night Transit and Safety Interventions

- Partner with late-night venue districts and rideshare/transit agencies to improve safe, accessible late-night transportation options, reducing impaired driving on Friday and Saturday nights.

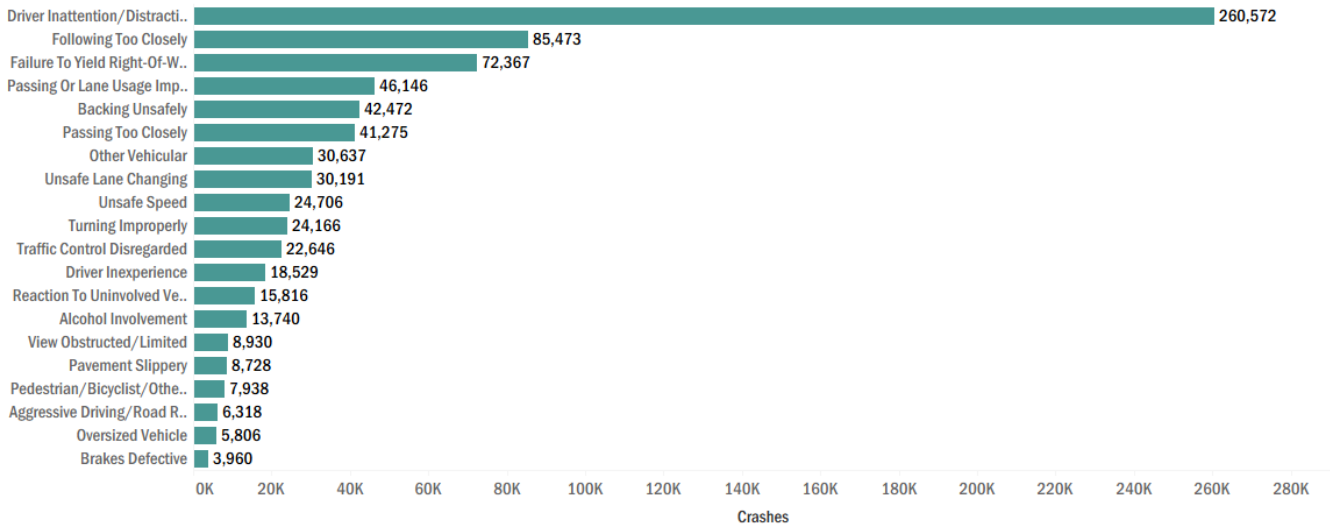
4. School Zone and Commercial Loading Enhancements

- Implement strict curbside loading regulations during peak afternoon hours (2:00 PM to 5:00 PM) to clear delivery trucks double-parking near schools and busy intersections.

CONTRIBUTING FACTORS ANALYSIS

Primary Reasons for Crashes

Out of nearly 800,000 crashes recorded, more than half (52%) were driven by just three human behaviors: driver inattention/distraction (260,572 crashes – 32.6%); following too closely/tailgating (85,473 crashes – 10.7%); and failure to yield right-of-way (72,367 crashes – 9%). DOT should run high-visibility public safety campaigns focusing specifically on the top two killers: mobile phone distraction and tailgating in heavy traffic.



Key Insights

Out of nearly 800,000 crashes where a specific cause was recorded, more than half (52%) were driven by just three human behaviors:

- 1. Driver Inattention/Distracti.. (260,572 crashes – 32.6%):**
 - By far the single biggest cause of crashes in NYC. Drivers looking at mobile phones, navigating, or otherwise taking their eyes off the road account for nearly 1 in every 3 accidents.
- 2. Following Too Closely/Tailgating (85,473 crashes – 10.7%):**
 - Drivers not leaving enough braking distance behind the car in front of them. When traffic suddenly slows down, this leads directly to rear-end collisions.
- 3. Failure to Yield Right-of-Way (72,367 crashes – 9.0%):**
 - Drivers failing to let pedestrians cross or cutting off oncoming vehicles when turning at intersections. This is one of the most dangerous causes for pedestrians and cyclists.
- 4. Improper Passing, Lane Usage and Unsafe Backing (88,618 crashes combined – 11.1%):**
 - Aggressive lane weaving, side-swiping while overtaking, and blind-spot accidents during reversing (especially involving delivery trucks and large vehicles).

5. Unsafe Speed, Disregarding Traffic Controls and Alcohol (61,092 crashes combined – 7.6%):

- Speeding, running red lights/stop signs, and drunk driving. While lower in total count compared to distraction, these behaviors carry an exceptionally high risk of severe injuries and fatalities.

Implications for NYC DOT

- **Human Error Dominates over Mechanical Failure:** Overwhelmingly, collisions in NYC are caused by driver behavior rather than vehicle malfunctions or poor road weather conditions.
- **Streets Must be Designed to Forgive Human Mistakes:** Since driver distraction cannot be eliminated entirely through warnings alone, street designs must force drivers to slow down and physically prevent fatal mistakes (e.g., traffic calming, protected zones).
- **Intersections are Critical Danger Zones:** Failure to yield, improper turning, and running red lights show that street intersections are where drivers and vulnerable road users (pedestrians and cyclists) conflict most violently.

Actionable Recommendations for DOT

1. Redesign Intersections to Enforce Yielding

- **Leading Pedestrian Intervals (LPis):** Program walk signals to give pedestrians a 3-7 second head start before vehicle traffic gets a green light, ensuring pedestrians are clearly visible in the crosswalk.
- **Curb Extensions (Bulb-Outs) and Hardened Centerlines:** Install rubber mounts or concrete curb extensions at turns. This forces drivers to make wider, slower turns rather than cutting quickly across crosswalks.

2. Automated Enforcement and Speed Management

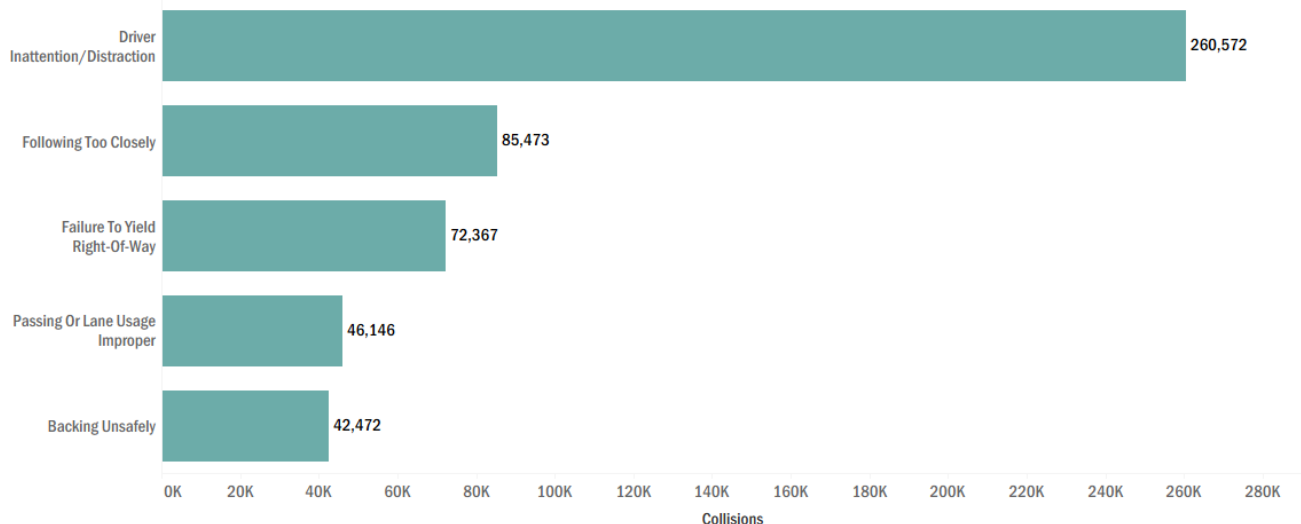
- **Expand Red-Light and Speed Camera Networks:** Deploy automated camera technology along high-injury corridors to consistently enforce speed limits and catch drivers running red lights without relying solely on police traffic stops.
- **Traffic Calming Measures:** Narrow overly wide driving lanes and install speed cushions on local streets to naturally encourage lower driving speeds and prevent tailgating.

3. Smart Technology and Distraction Campaigns

- **In-Vehicle Telematics for Fleet Vehicles:** Require NYC municipal vehicles and private commercial trucks/taxis to install driver-assistance technologies (e.g., automatic emergency braking and forward-collision warning systems) that counteract tailgating and distraction.
- **Targeted Public Awareness:** Run high-visibility public safety campaigns focusing specifically on the top two killers: mobile phone distraction and tailgating in heavy traffic.

Top 5 Reasons for Crashes

The top 5 recorded causes account for **507,030 total collisions**. Human behavior drives the vast majority of risk. DOT should expand Neighborhood Loading Zones (NLZs) for commercial trucks to reduce double-parking and unsafe reversing maneuvers on busy avenues.



The top 5 recorded causes account for **507,030 total collisions**:

Driver Inattention/Distracted driving

(260,572 crashes – 51.39% of top 5). More than half of all top 5 crashes are caused by distracted driving. This includes drivers looking at smartphones, adjusting navigation systems, or taking their attention off the roadway.

Following Too Closely/Tailgating

Failure to Yield Right-of-Way

(72,367 crashes – 14.27% of top 5). Drivers fail to let pedestrians cross or cut off oncoming traffic when turning at intersections. This is one of the most hazardous behaviors for pedestrians and cyclists.

Passing or Lane Usage Improper

(85,473 crashes – 16.86% of top 5). Tailgating is the second most common factor. Drivers do not leave enough stopping distance, leading directly to rear-end collisions when traffic slows down suddenly.

(46,146 crashes – 9.10% of top 5). Reckless merging, aggressive lane weaving, and unsafe overtaking cause sideswipe collisions along multi-lane avenues and highways.

Backing Unsafely

(42,472 crashes – 8.38% of top 5). Blind-spot collisions occurring while vehicles—especially delivery trucks, commercial vans, and SUVs—reversely maneuver in dense parking or loading zones.

Key Implications for NYC DOT

- **Human Behavior Drives the Vast Majority of Risk:** The overwhelming majority of top crashes stem from preventable driver decisions rather than road surface conditions or mechanical failures.
- **Roads Must Force Slower, Safer Driving:** Because warnings alone do not stop distraction or tailgating, street geometry must physically limit vehicle speeds and reduce conflict points.
- **Commercial and Loading Friction:** Unsafe backing and improper passing frequently involve delivery vehicles blocking active travel lanes, creating blind spots for other motorists.

Actionable Recommendations for DOT

1. Intersection Safety Overhauls (Failure to Yield and Passing)

- **Leading Pedestrian Intervals (LPis):** Program walk signals to give pedestrians a 3- to 7-second head start before turning vehicle traffic gets a green light.
- **Hardened Centerlines and Raised Crosswalks:** Install rubber mounts or concrete curb extensions at turns to force drivers to make wider, slower turns instead of cutting across crosswalks.

2. Automated Enforcement and Speed Management (Tailgating and Distraction)

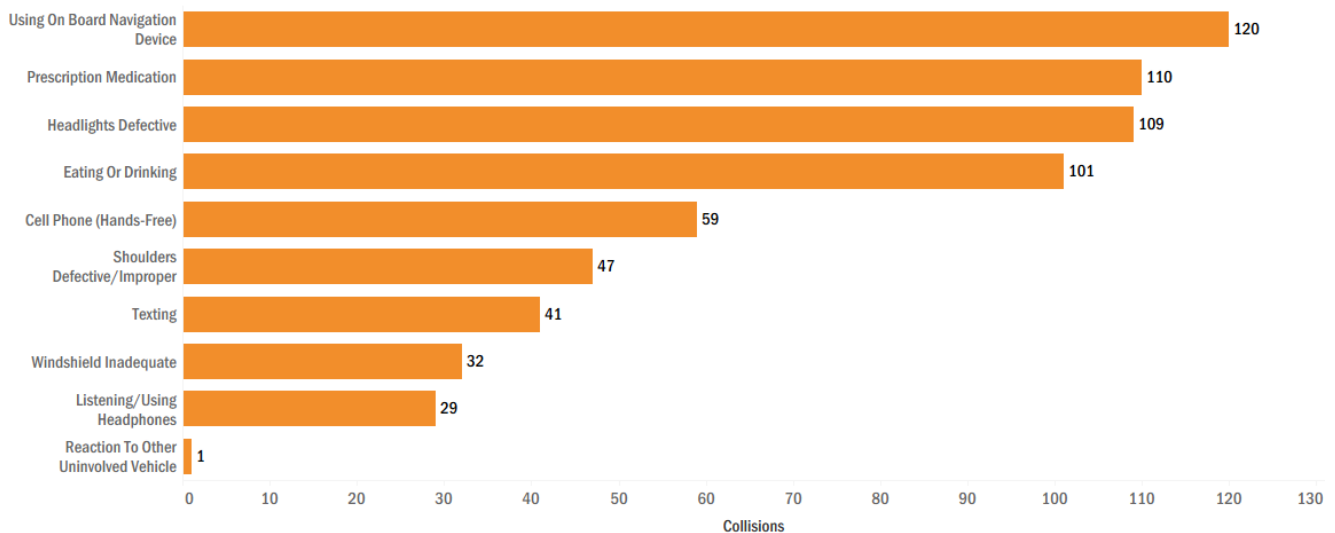
- **Expand Camera Safety Networks:** Deploy red-light cameras and automated speed enforcement along high-injury corridors to deter tailgating and aggressive driving.
- **Traffic Calming Infrastructure:** Narrow overly wide driving lanes and add speed cushions on local streets to naturally encourage lower driving speeds and safe trailing distances.

3. Curbside and Delivery Management (Unsafe Backing)

- **Dedicated Loading Zones:** Expand Neighborhood Loading Zones (NLZs) for commercial trucks to reduce double-parking and unsafe reversing maneuvers on busy avenues.
- **Fleet Safety Standards:** Require city contractor vehicles and commercial fleets to utilize proximity sensors, backup cameras, and forward-collision warning technology.

The Least Contributing Factors to Crashes

The low prevalence of mechanical failures shows that vehicle safety inspections and standards are effective. DOT should collaborate with the NYPD to update digital collision reporting software so officers can easily check sub-categories of distraction (e.g., handheld phone vs. texting) without defaulting entirely to generic "Inattention."



Looking at the data, several specific factors appear with extremely low crash counts:

- **Ultra-Low Recorded Behaviors and Minor Distractions:**
 - *Listening/Using Headphones* (29 crashes)
 - *Texting* (41 crashes)
 - *Cell Phone (Hands-Free)* (59 crashes)
 - *Eating or Drinking* (101 crashes)
 - *Using On-Board Navigation Devices* (120 crashes)
- **Vehicle Equipment and Mechanical Defects:**
 - *Windshield Inadequate* (32 crashes)

- *Headlights Defective* (109 crashes)
- *Tow Hitch Defective* (133 crashes)
- *Other Lighting Defects* (145 crashes)
- *Tinted Windows* (177 crashes)

The Real Story Behind the Low Numbers

At first glance, it might look like texting or using cell phones rarely causes crashes in NYC.

However, **this is a reporting artifact rather than reality:**

1. **The "Catch-All" Effect:** When police officers file crash reports at the scene, unless a driver explicitly admits to texting or using headphones, officers routinely log the cause under the broad catch-all category **"Driver Inattention/Distraction"** (which accounts for over **315,000 crashes** at the very top of the table).
2. **Vehicle Defects Are Genuinely Rare:** Modern safety standards mean that mechanical issues like defective windshields or tow hitches truly contribute to less than 0.1% of all crashes in New York City.

Implications for NYC DOT

- **Significant Underreporting of Specific Distractions:** Relying purely on specific category counts (like "Texting = 41") drastically underestimates how smartphone use impacts NYC traffic safety.
- **Vehicle Inspection Rules are Working:** The low prevalence of mechanical failures shows that vehicle safety inspections and standards are effective, meaning physical equipment failures are not a primary driver of city crashes.
- **Prioritization of Safety Budget:** The DOT should avoid allocating major resources toward low-impact physical defects or minor niche causes, maintaining focus on high-impact street redesigns and general speed enforcement.

Recommendations for DOT

1. **Modernize Crash Data Collection with NYPD**

- **Refine Reporting Forms:** Collaborate with the NYPD to update digital collision reporting software so officers can easily check sub-categories of distraction (e.g., handheld phone vs. texting) without defaulting entirely to generic "Inattention."
- **Digital and Telematics Integration:** Utilize anonymized cellular and connected vehicle telematics data to gain truer insights into where phone interaction occurs most frequently along high-risk corridors.

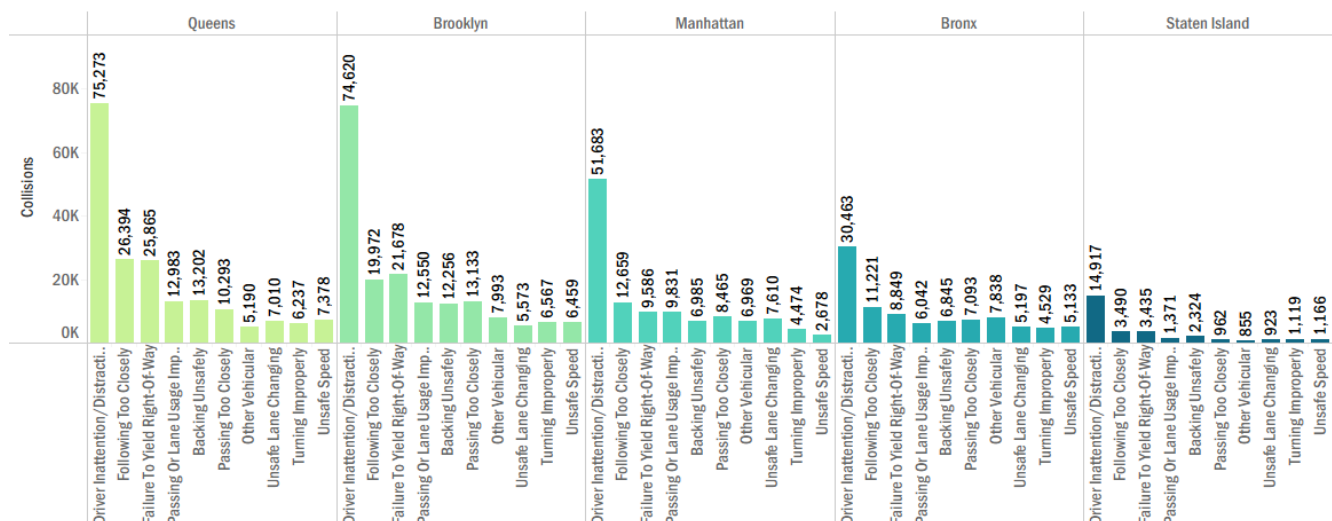
2. Maintain Focus on Core Behavioral Interventions

- **Broad Anti-Distracted Enforcement:** Continue public safety campaigns targeting all forms of phone interaction and distracted driving rather than focusing narrowly on specific devices or hands-free tech.
- **Infrastructure That Counteracts Distraction:** Since driver inattention will always exist to some degree, continue installing physical traffic-calming measures—such as raised crosswalks, protected bike lanes, and speed cushions—that physically slow vehicles down regardless of driver focus.

Top 10 Ranked Primary Reason for Crashes by Borough

An analysis of the top 10 crash causes across New York City's five boroughs reveals distinct patterns driven by local geography, road designs, and traffic flow.

Speeding (Unsafe Speed) is significantly more common in outer boroughs like **Queens (7,378 crashes, Rank #7)**, **Brooklyn (6,459 crashes, Rank #10)**, **Bronx (5,133 crashes, Rank #9)**, and **Staten Island (1,166 crashes, Rank #6)**. DOT should expand automated speed camera coverage along major multi-lane corridors in Queens, Brooklyn, Bronx, and Staten Island where speeding ranks in the top 10.



Key Comparisons Between Boroughs

1. Driver Distraction is the Universal #1 Threat

- **Driver Inattention/Distraction** is the top-ranked cause of crashes across **all five boroughs**.
- It accounts for the single highest volume of collisions in Queens (75,273 crashes), closely followed by Brooklyn (74,620 crashes) and Manhattan (51,683 crashes).

2. Speeding: Outer Boroughs vs. Manhattan

- **Speeding (Unsafe Speed)** is significantly more common in outer boroughs like **Queens** (7,378 crashes, Rank #7), **Brooklyn** (6,459 crashes, Rank #10), **Bronx** (5,133 crashes, Rank #9), and **Staten Island** (1,166 crashes, Rank #6).
- In **Manhattan**, **Unsafe Speed does not even appear in the top 10**. Dense traffic gridlock and lower speed limits naturally keep vehicle speeds down, whereas wide multi-lane highways in Queens and Brooklyn encourage faster driving.

3. Intersection Violations (Failure to Yield and Running Signals)

- **Failure to Yield Right-of-Way** is particularly high in **Brooklyn** (Rank #2, 21,678 crashes) and **Queens** (Rank #3, 25,865 crashes), where dense pedestrian crossings intersect with heavy vehicle traffic.
- **Traffic Control Disregarded** (running red lights and stop signs) makes the top 10 list specifically in **Brooklyn** (7,039 crashes, Rank #8) and **Queens** (6,663 crashes, Rank #9).

4. Tailgating and Merge Conflicts

- **Following Too Closely** is the **#2 primary factor** in Queens (26,394 crashes), Manhattan (12,659 crashes), Bronx (11,221 crashes), and Staten Island (3,490 crashes), and **#3** in Brooklyn (19,972 crashes).
- **Improper Passing and Lane Changing** rank higher in **Manhattan** (Rank #3) compared to other boroughs, reflecting frequent maneuvering around double-parked delivery trucks and yellow taxis.

Implications for NYC DOT

- **A "One-Size-Fits-All" Approach Will Not Work:** Manhattan requires street design policies focused on managing curb space and tight-lane maneuvering, whereas outer boroughs like Queens, Brooklyn, and Staten Island need interventions focused on speed control and arterial corridors.

- **Intersection Conflicts in High-Density Boroughs:** The high rank of "Failure to Yield" in Brooklyn and Queens highlights that turning vehicles pose severe threats to pedestrians and cyclists at wide arterial junctions.

Recommendations for DOT

1. Borough-Specific Speed Enforcement (Queens, Brooklyn, Bronx, Staten Island)

- Expand automated speed camera coverage along major multi-lane corridors in Queens, Brooklyn, Bronx, and Staten Island where speeding ranks in the top 10.
- Implement "road diets" (reducing lane width or reallocating lanes to bus/bike paths) on wide avenues to physically prevent speeding.

2. Manhattan Curb and Lane Management

- Create dedicated **Commercial Loading Zones** to reduce double-parking on narrow Manhattan streets, minimizing the risky lane-weaving and improper passing that drives Manhattan crashes.

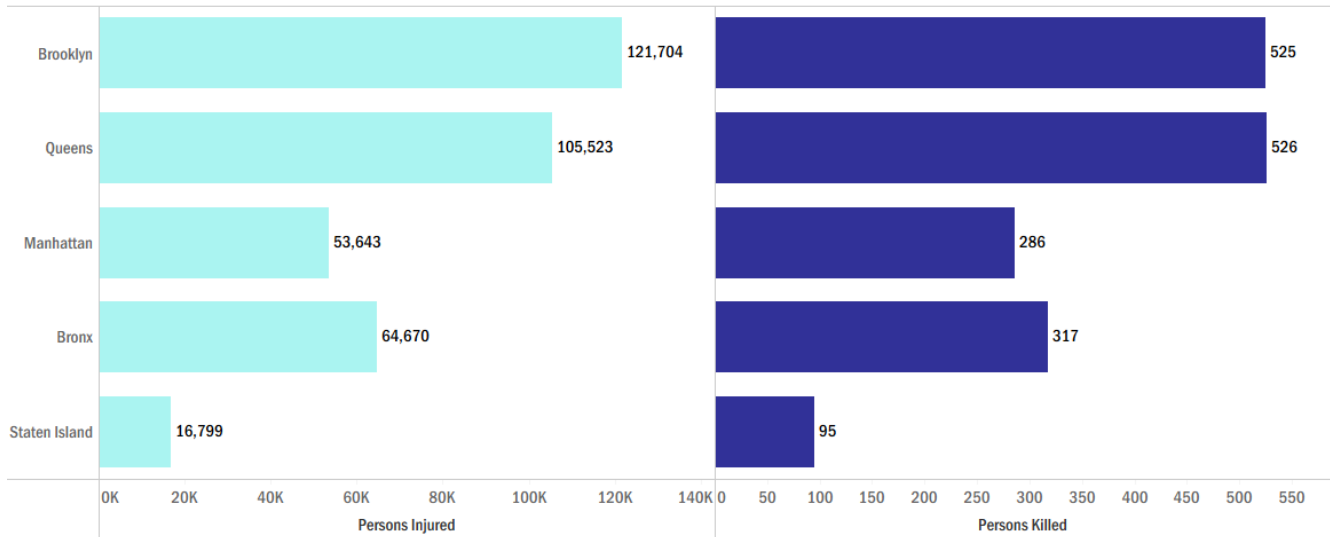
3. Targeted Signal Infrastructure in Brooklyn and Queens

- Install **Leading Pedestrian Intervals (LPIs)** and dedicated left-turn signal phases at top intersections in Brooklyn and Queens to force vehicles to yield right-of-way.

INJURY AND FATALITY ANALYSIS

Total Number of People Injured and Killed by Borough

Brooklyn and Queens suffer the highest toll (63% of all casualties). Brooklyn tops the city in injuries with **121,704 people injured** and **525 fatalities**. DOT should Focus Vision Zero street redesign capital projects primarily on known high-injury corridors in Brooklyn and Queens (e.g., Atlantic Avenue, Flatbush Avenue Extension, Linden Boulevard).



Breakdown of Casualties by Borough

Across all reported collisions, over 360,000 injuries and 1,749 fatalities were recorded:

- **Brooklyn and Queens Suffer the Highest Toll (63% of All Casualties):**
 - Brooklyn tops the city in injuries with **121,704 people injured** and **525 fatalities**.
 - Queens has nearly identical fatality numbers (**526 killed**) and recorded **105,523 injuries**.
 - Combined, these two boroughs account for **nearly two-thirds of all crash-related injuries and deaths in NYC**.
- **The Bronx (18% of Casualties):**
 - Recorded **64,670 injuries** and **317 fatalities**. While lower than Brooklyn and Queens, the Bronx has a higher severity rate relative to its total crash count due to major highway/arterial traffic flows.
- **Manhattan (15% of Casualties):**
 - Recorded **53,643 injuries** and **286 fatalities**. Lower overall speeds caused by dense traffic gridlock help keep fatal crash rates lower than outer boroughs, though high pedestrian volume keeps injury numbers significant.
- **Staten Island (5% of Casualties):**

- Recorded **16,799 injuries** and **95 fatalities**. Lower population density and overall crash volume keep Staten Island's absolute numbers the lowest in the city.

Implications for NYC DOT

- **Outer Borough Avenues are Lethal Danger Zones:** High casualty counts in Brooklyn, Queens, and the Bronx are heavily driven by wide, multi-lane arterial avenues (such as Atlantic Ave, Linden Blvd, and Northern Blvd) where higher vehicle speeds directly translate to more severe injuries and higher fatality rates.
- **Resource Allocation Must Prioritize Brooklyn and Queens:** Because over 60% of all traffic injuries and deaths occur in Brooklyn and Queens, DOT safety budgets, physical street overhauls, and police enforcement initiatives should be weighted heavily toward these two boroughs.
- **High Collision Volume Doesn't Always Equal High Fatalities:** Manhattan has massive vehicle and pedestrian activity, but gridlock limits vehicle speeds. Conversely, outer borough roads allow higher speeds, making crashes far more deadly when they happen.

Recommendations for DOT

1. Prioritize Safety Interventions in Brooklyn and Queens

- Focus Vision Zero street redesign capital projects primarily on known high-injury corridors in Brooklyn and Queens (e.g., Atlantic Avenue, Flatbush Avenue Extension, Linden Boulevard).

2. Target High-Speed Arterials with "Road Diets"

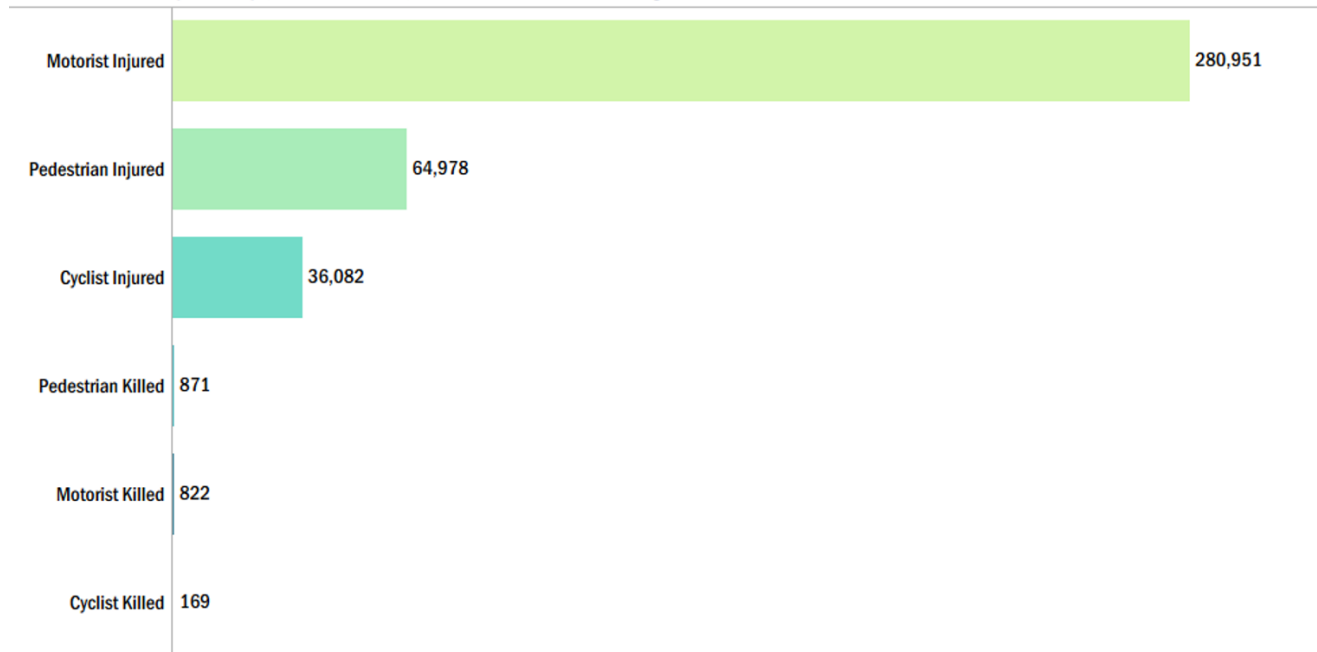
- Re-engineer wide multi-lane roads in outer boroughs by narrowing lane widths, adding pedestrian refuge islands, and reallocating road space to dedicated bus or bike lanes to naturally slow down traffic.

3. Protect Vulnerable Pedestrians at Intersections

- Expand **Leading Pedestrian Intervals (LPIs)** and dedicated left-turn signals across major Brooklyn and Queens intersections to give pedestrians head starts when crossing and reduce "Failure to Yield" turning accidents.

Injuries and Fatalities by Mode of Transport

A pedestrian involved in a crash is over 4 times more likely to die compared to a motorist in a crash (1.32% fatality rate vs. 0.29%). DOT should implement street-calming interventions such as speed humps, raised crosswalks, and narrower vehicle lanes on high-incident corridors.



Key Findings Summary

- **Total Injuries:** 382,011 across all groups.
- **Total Fatalities:** 1,862 lives lost.
- **Total Casualties:** 383,873 combined.

Which Groups Are Most Affected?

1. Motorists Experience the Highest Volume of Casualties

- 73.5% of all injuries (280,951) involve drivers and car passengers.
- Motorists account for 44.1% of all fatalities (822 deaths).
- **Why:** High speeds, vehicle-to-vehicle impacts, and high traffic volume contribute to frequent, non-fatal crashes.

2. Pedestrians Face the Highest Risk of Death

- Pedestrians account for 46.8% of all deaths (871 fatalities), making them the group most likely to suffer fatal injuries in a crash.
- A pedestrian involved in a crash is over 4 times more likely to die compared to a motorist in a crash (1.32% fatality rate vs. 0.29%).

- **Why:** Pedestrians have zero physical protection when struck by vehicles, especially heavy trucks or SUVs.

3. Cyclists Face High Vulnerability Relative to Infrastructure

- Cyclists account for **36,082 injuries** and **169 fatalities**.
- While total numbers are lower, cyclists share similar physical exposure to pedestrians when sharing lanes with motor vehicles.

Key Implications for NYC DOT

- **Shift Focus to Vulnerable Road Users (VRUs):** While drivers suffer the majority of injuries, pedestrians suffer the majority of deaths. Urban design policies must prioritize protecting people outside of cars.
- **Speed is the Primary Hazard:** The high lethality rate for pedestrians points to vehicle speed as a critical factor in fatal impacts.
- **Street Design Deficits:** Current street layouts still allow dangerous interactions where fast-moving vehicles mix with walking and cycling paths.

Recommendations for DOT

1. Expand Protected Pedestrian Zones and Signal Timings:

- Increase **Leading Pedestrian Intervals (LPIs)** at busy intersections to give walkers a head start before cars can turn.
- Install sidewalk extensions (curb bulb-outs) to shorten crossing distances.

2. Build Fully Protected Bike Networks:

- Transition from painted bike lanes to **physically separated, barrier-protected bike lanes** to prevent vehicle encroachment and keep cyclists safe.

3. Lower Speed Limits and Enforce Calming Measures:

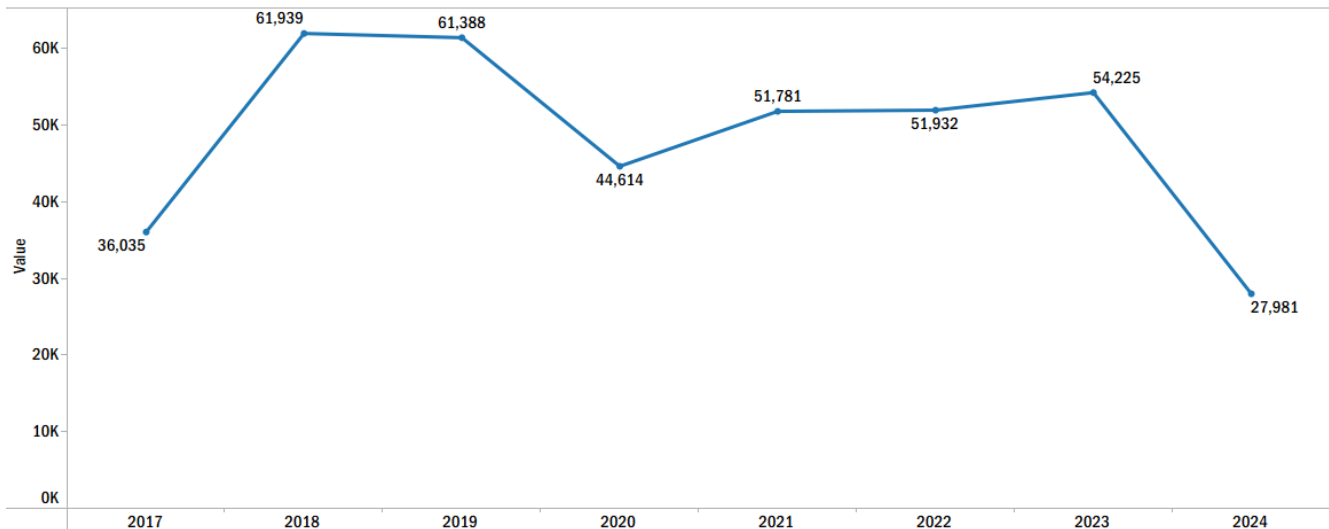
- Implement street-calming interventions such as speed humps, raised crosswalks, and narrower vehicle lanes on high-incident corridors.

4. Redesign Danger-Prone Intersections:

- Target top crash intersections with daylighting (removing parking spots near corners to improve visibility) and restricted left-turn movements.

Injury Trends Over Time

Injuries peaked at **61,939** in 2018 and remained high at **61,388** in 2019. Injuries dropped significantly by **27.3%** to **44,614** in 2020. This decline corresponds to COVID-19 lockdowns and reduced vehicle travel. As traffic returned, injuries began climbing steadily back up, reaching **51,781** in 2021, **51,932** in 2022, and **54,225** in 2023. DOT should deploy speed and red-light cameras in areas showing post-pandemic increases in collisions.



Findings Summary

An analysis of annual crash injuries reveals clear phases in traffic safety over the reported years:

- Pre-Pandemic Peak (2018-2019):** Injuries peaked at **61,939** in 2018 and remained high at **61,388** in 2019.
- Pandemic Drop (2020):** Injuries dropped significantly by **27.3%** to **44,614** in 2020. This decline corresponds to COVID-19 lockdowns and reduced vehicle travel.
- Post-Pandemic Rebound (2021-2023):** As traffic returned, injuries began climbing steadily back up, reaching **51,781** in 2021, **51,932** in 2022, and **54,225** in 2023.
- Partial Year Data (2017 & 2024):**
 - 2017 (**36,035**) and 2024 (**27,981**) reflect partial-year data collection periods rather than sudden drops in overall safety.

Implications for NYC DOT

- Return to Pre-Pandemic Risks:** The steady rise in injuries from 2021 through 2023 indicates that safety risks have returned alongside normal traffic volumes.
- Temporary Changes Don't Fix Structural Risks:** Reduced road usage during 2020 temporarily lowered injury numbers, but without permanent infrastructure changes, injury rates bounce back as traffic grows.

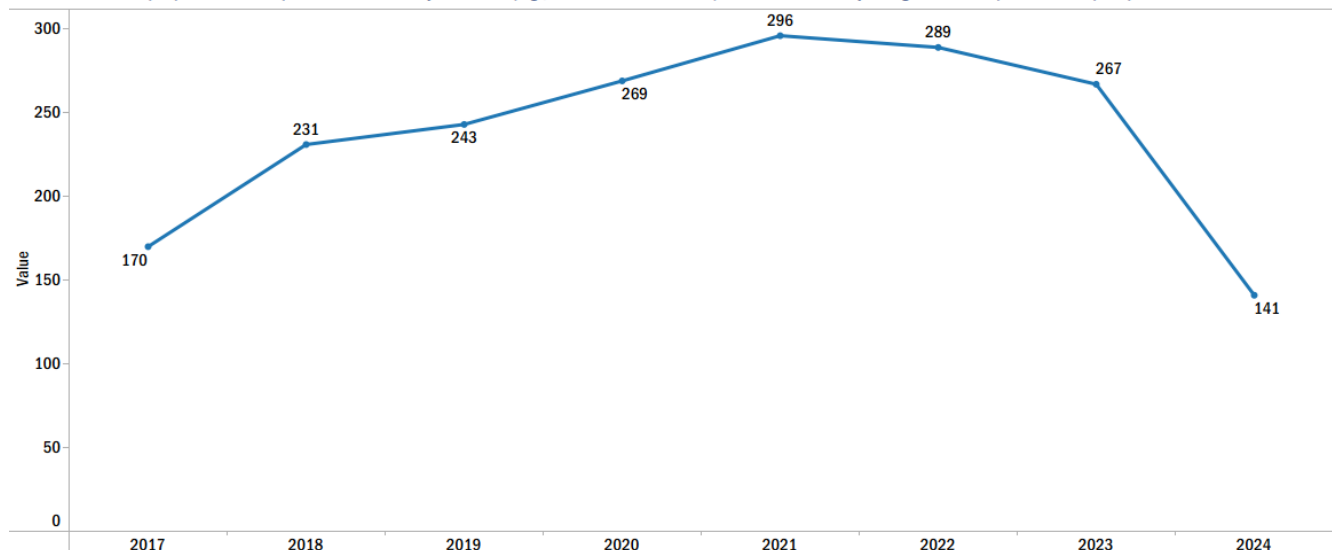
- **Re-evaluating Vision Zero Strategies:** A multi-year baseline averaging over 51,000 injuries annually post-2020 highlights the need for targeted interventions on high-risk corridors.

Key Recommendations

1. **Prioritize Targeted Safety Interventions:** Focus capital projects on high-volume crash corridors that drove the 2021-2023 rebound.
2. **Expand Automated Enforcement:** Deploy speed and red-light cameras in areas showing post-pandemic increases in collisions.
3. **Redesign High-Risk Intersections:** Implement physical traffic calming (such as raised crosswalks, protected turns, and curb extensions) to permanently reduce crash severity regardless of traffic volume.

Fatality Trends Over Time

Traffic deaths rose steadily from 231 in 2018 to a peak of 296 in 2021. Since 2021, fatalities began to cool down, dropping to 289 in 2022, 267 in 2023, and 141 in 2024. DOT should advocate for state and municipal policies that improve vehicle visibility standards (e.g., direct-vision truck cabs) and introduce safety ratings tailored to pedestrian impact protection.



Findings Interpretation:

1. **Fatalities Peaked in 2021:**
 - Traffic deaths rose steadily from 231 in 2018 to a peak of 296 in 2021.
 - Since 2021, fatalities have begun to cool down, dropping to 289 in 2022, 267 in 2023, and 141 in 2024 (partial year).

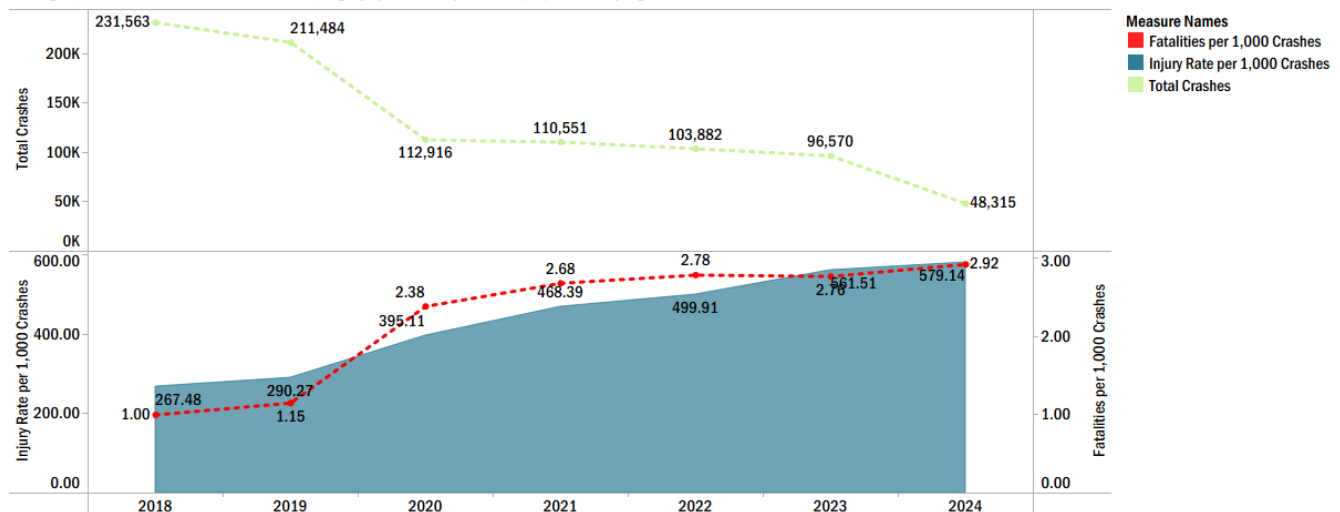
2. **The Disparity: Why Did Fatalities Rise While Crashes Fall?** While total crashes dropped by 58.3% between 2018 and 2023 (from 231,563 down to 96,570), fatalities went *up* by 15.6% over that same timeframe. Three core real-world factors explain this disparity:

- **Increased Crash Severity (Higher Speeds):** During the 2020 lockdowns, traffic congestion dramatically decreased. Open roadways allowed drivers to travel at significantly higher speeds. Because collision force increases quadratically with speed, fewer overall crashes occurred, but those that did happen were far more deadly.
- **Shift in Vehicle Types (Larger Vehicles):** The proportion of heavier vehicles – such as SUVs, pickup trucks, and large delivery vans – on city streets increased. Heavier vehicles exert substantially greater kinetic energy in a collision, making a crash much more likely to turn fatal for pedestrians and cyclists.
- **Minor Fender-Benders Went Unreported:** The sharp drop in total crashes (from 230k+ to 100k) was largely driven by a reduction in minor, low-speed bumper crashes that no longer required police reports. However, severe and high-speed crashes continued to occur, keeping injury and fatality totals disproportionately high relative to total recorded crashes.

Visual Comparison: The Disparity at a Glance

Total crashes fell sharply from 2018 to 2020, while fatalities per 1,000 crashes nearly tripled from 1.0 per 1,000 in 2018 to 2.78 per 1,000 in 2022.

While total crashes dropped by 58.3% between 2018 and 2023 (from 231,563 down to 96,570), fatalities went *up* by 15.6% over that same timeframe. The sharp drop in total crashes (from 230k+ to ~100k) was largely driven by a reduction in minor, low-speed bumper crashes that no longer required police reports. However, severe and high-speed crashes continued to occur, keeping injury and fatality totals disproportionately high relative to total recorded crashes.



Implications for NYC DOT

- **Crash Count is a Misleading Metric for Safety:** Relying solely on total crash volume creates a false sense of security. Fewer fender-benders do not equate to safer streets if severe outcomes per collision are rising.
- **Speed Management Remains the Primary Safety Lever:** The dramatic jump in fatalities per crash highlights that impact velocity – not just crash frequency – is the determining factor in whether victims survive.
- **Vulnerable Road Users Suffer Most from Heavy Vehicles:** As vehicle size increases, pedestrians and cyclists bear the brunt of severe collisions.

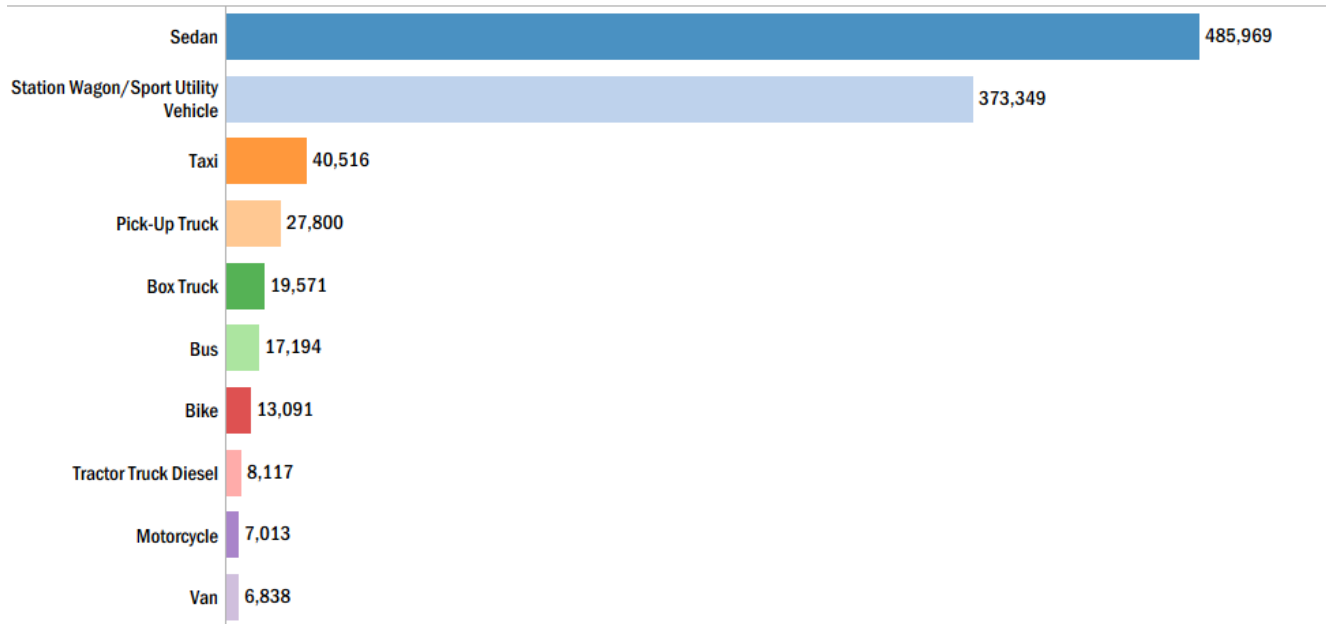
Key Recommendations

1. **Traffic Calming and Street Narrowing:** Redesign wide, open avenues with physical speed reducers (raised crosswalks, speed cushions, and narrower driving lanes) to prevent high-speed driving even during low-traffic hours.
2. **Expand Automated Speed Enforcement:** Maintain continuous 24/7 speed camera operations across high-risk corridors to deter reckless driving.
3. **Vehicle Safety and Size Regulations:** Advocate for state and municipal policies that improve vehicle visibility standards (e.g., direct-vision truck cabs) and introduce safety ratings tailored to pedestrian impact protection.
4. **Daylighting Intersections:** Clear parking spaces adjacent to crosswalks to widen lines of sight for drivers turning near pedestrians and cyclists.

VEHICLE TYPES ANALYSIS

Vehicle Types and Crashes

Everyday personal passenger cars drive the vast majority of collisions on NYC streets. Sedans (48.6%) and Station Wagons/SUVs (37.4%) account for 86% of all recorded crashes combined (859,318 total incidents). DOT should continue building physically protected bike lanes to isolate cyclists from the massive volume of sedans and SUVs on shared streets.



Findings Interpretation

An analysis of crashes categorized by vehicle type highlights the following breakdown:

1. Passenger Vehicles Dominate Crash Numbers:

- Sedans (48.6%) and Station Wagons/SUVs (37.4%) account for 86% of all recorded crashes combined (859,318 total incidents).
- Everyday personal passenger cars drive the vast majority of collisions on NYC streets.

2. Commercial and Service Vehicles Form a Secondary Tier:

- Taxis (4.1%), Pick-Up Trucks (2.8%), and Box Trucks (2.0%) represent the largest non-passenger vehicle categories involved in crashes.

3. Vulnerable Modes Appear Less Frequently in Total Volume:

- Bikes (1.3%) and Motorcycles (0.7%) account for a small fraction of overall collisions. However, when these smaller vehicles are involved, outcomes tend to be disproportionately severe.

Implications for NYC DOT

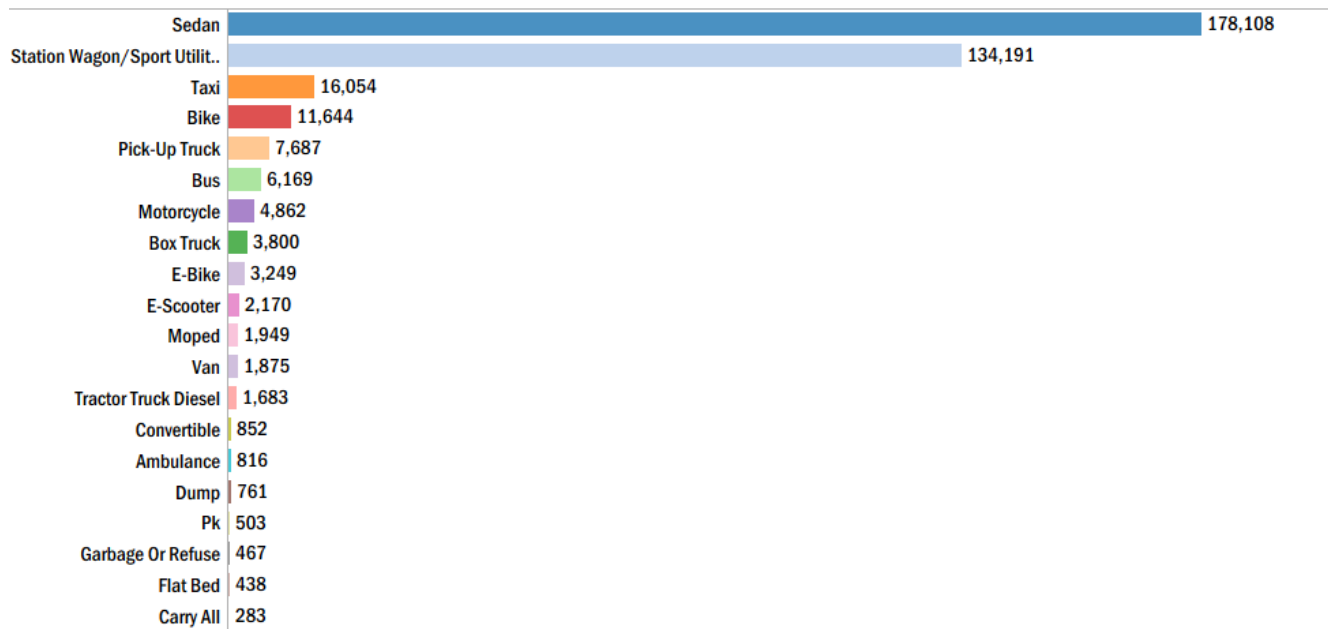
- **Focus Strategy Beyond Commercial Fleets:** While heavy trucks and taxis often draw public attention, personal passenger vehicles (sedans and SUVs) are responsible for over 8 out of every 10 crashes.
- **The Rise of SUVs Increases Risk:** SUVs and Station Wagons alone account for over 373,000 crashes. Larger, heavier vehicles pose higher risks of serious injury to pedestrians and cyclists due to higher front grilles and blind spots.
- **Commercial Loading Conflicts:** Box trucks and pickup trucks contribute over 47,000 crashes, often stemming from double parking, loading zone congestion, and tight turns on narrow streets.

Key Recommendations

1. **Target Personal Passenger Driver Behavior:** Expand automated speed and red-light camera enforcement in residential and commercial corridors where sedan and SUV traffic is concentrated.
2. **Redesign Loading Zones for Delivery Trucks:** Establish dedicated curb loading zones for box trucks and pickup trucks to reduce double-parking hazards and side-swipe collisions on busy avenues.
3. **Enhance Direct-Vision Standards and Vehicle Safety Advocacy:** Advocate for safety standards that improve driver visibility in large SUVs and light trucks to reduce blind-spot impacts with pedestrians and cyclists.
4. **Physically Separate Vulnerable Road Users:** Continue building physically protected bike lanes to isolate cyclists from the massive volume of sedans and SUVs on shared streets.

Vehicle Types and Injury

Emerging micromobility and two-wheelers show growing injury footprints. Traditional Bikes (11,644 injuries, 3.2%), E-Bikes (3,249 injuries, 0.9%), and E-Scooters (2,170 injuries, 0.6%) collectively account for over 17,000 injuries (4.6%). Motorcycles account for 4,862 injuries (1.3%). DOT should partner with the Taxi and Limousine Commission (TLC) to enforce safe passenger drop-off practices and prevent "dooring" incidents that injure cyclists.



Key Findings Summary

An analysis of injury data categorized by vehicle type reveals the following breakdown:

1. Passenger Vehicles Account for Most Injuries:

- Sedans (48.4%) and Station Wagons / SUVs (36.5%) account for 84.9% of all injuries combined (312,299 total injuries).
- The vast majority of injuries on NYC streets involve standard personal cars.

2. Taxis and For-Hire Vehicles Represent the Next Largest Group:

- Taxis account for 16,054 injuries (4.4%), making them the third highest individual category.

3. Emerging Micromobility and Two-Wheelers Show Growing Injury Footprints:

- Traditional Bikes (11,644 injuries, 3.2%), E-Bikes (3,249 injuries, 0.9%), and E-Scooters (2,170 injuries, 0.6%) collectively account for over 17,000 injuries (4.6%).
- Motorcycles account for 4,862 injuries (1.3%).

Key Implications for NYC DOT

- **Personal Passenger Cars Are the Primary Threat to Street Safety:** While commercial vehicles and micro-mobility often dominate public discussion, standard sedans and SUVs cause nearly 85 out of every 100 injuries.

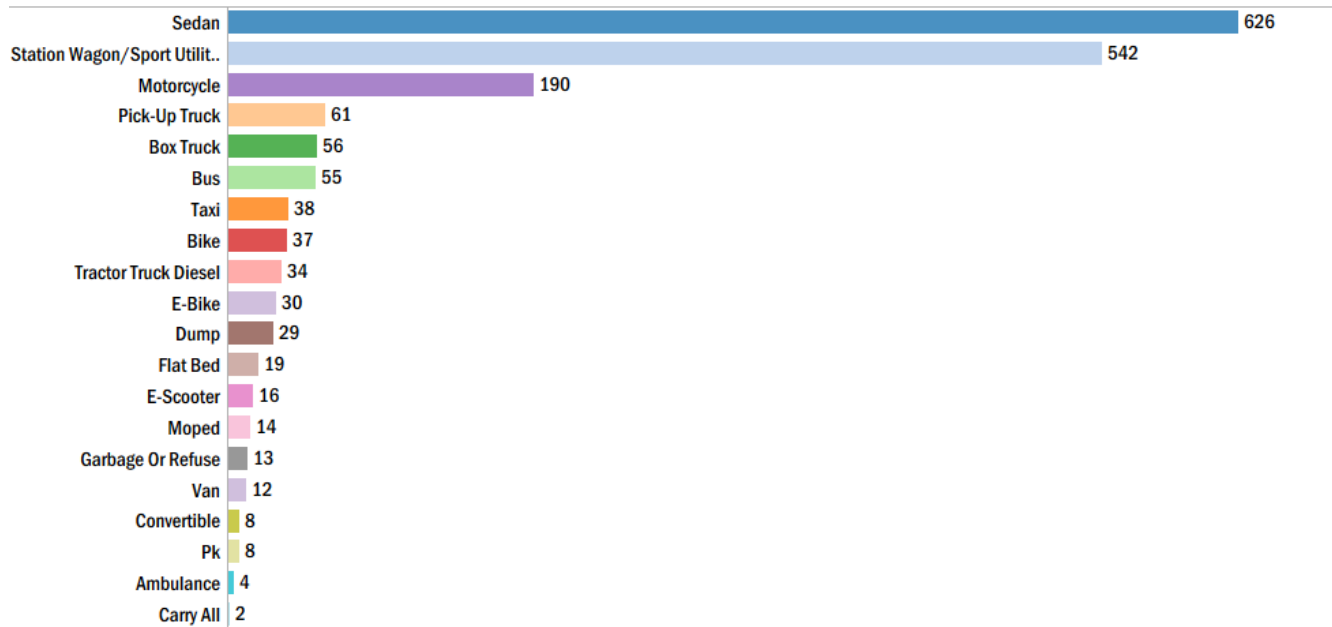
- **High Injury Rate for Two-Wheelers Relative to Volume:** Traditional bikes, e-bikes, e-scooters, and motorcycles experience significant injury numbers despite making up a small fraction of overall vehicles on the road.
- **Rising Complexity of Street Use:** The presence of e-bikes and e-scooters alongside traditional bicycles requires redesigned curb space and lane separations to manage varying travel speeds.

Key Recommendations

1. **Focus Traffic Calming on Passenger Car Corridors:** Deploy street-calming infrastructure—such as speed humps, raised crosswalks, and narrower vehicle lanes—on major avenues heavily trafficked by sedans and SUVs.
2. **Expand Protected Bike and Micromobility Infrastructure:** Build wide, physically barrier-protected lanes dedicated to bicycles, e-bikes, and e-scooters to reduce conflicts with motor vehicles.
3. **Enhance Taxi Driver Safety Programs:** Partner with the Taxi and Limousine Commission (TLC) to enforce safe passenger drop-off practices and prevent "dooring" incidents that injure cyclists.
4. **Daylighting Intersections:** Remove parking spaces closest to corners across high-risk corridors to increase visibility between drivers, pedestrians, and two-wheeled riders.

Vehicle Types and Fatality

Combined commercial/heavy vehicles (**Trucks and Heavy Vehicles**) cause a substantial portion of severe and fatal impacts (**209 fatalities, 11.5%**). DOT should require side guards, blind-spot mirrors, and high-visibility cab requirements on commercial box trucks, freight diesel trucks, and municipal garbage trucks operating in dense pedestrian zones.



Key Summary

Vehicle Category	Fatalities	Share of Total Fatalities	Key Insight
Sedans and Passenger Cars	626	34.5%	The single largest contributor to traffic deaths across the city.
SUVs and Station Wagons	542	29.9%	The second largest contributor; combined with sedans, standard passenger cars accounts for nearly two-thirds (64.5%) of all fatalities.
Motorcycles	190	10.5%	Overrepresented in fatal crashes relative to their small share of overall road traffic.
Trucks and Heavy Vehicles (Box, Pick-Up, Diesel, Dump, Refuse)	209	11.5%	Combined commercial/heavy vehicles cause a substantial portion of severe and fatal impacts.
Buses & Taxis	93	5.1%	Public transit and ride-hailing/fleet vehicles account for a smaller, distinct portion of fatalities.
Micro-mobility and Bicycles (Bikes, E-Bikes, E-Scooters, Mopeds)	113	6.2%	Represents fatalities involving light two-wheeled devices.

Implications for NYC DOT

- Standard Cars Are the Primary Threat:** Over **64% of traffic deaths** involve everyday passenger cars (sedans, SUVs, and station wagons). While public debate often focuses on delivery workers or e-scooters, standard motor vehicles remain the primary source of fatal traffic violence in New York City.

- **Vehicle Size and Weight Escalate Fatal Risk:** SUVs and light trucks account for over 600 deaths combined. Higher hood heights, larger blind spots, and greater vehicle weight significantly increase the likelihood that a collision with a pedestrian or cyclist becomes fatal rather than non-fatal.
- **High Vulnerability of Two-Wheel Riders:** Motorcycles alone account for over 10% of deaths (190 fatalities), showing that high-speed motorized two-wheelers present an exceptionally high risk of fatal injury compared to their low volume on city streets.

Recommendations for NYC DOT

1. **Redesign Streets for Lower Vehicle Speeds:**
 - **Target High-Volume Arterials:** Focus street redesigns (lane narrowings, speed cushions, daylighted intersections) on corridors dominated by sedan and SUV traffic, where lower speeds directly reduce fatal impacts.
2. **Address Large Vehicle Safety and Blind Spots:**
 - **Mandate Safety Guards & Direct Vision Standards:** Require side guards, blind-spot mirrors, and high-visibility cab requirements on commercial box trucks, freight diesel trucks, and municipal garbage trucks operating in dense pedestrian zones.
3. **Expand Dedicated and Physically Protected Infrastructure:**
 - **Hard Barriers for Micro-Mobility:** Replace painted bike lanes with concrete-bollard or curb-protected lanes to shield non-motorized and light electric vehicles from passenger car traffic.
4. **Enhance Motorcycle and Speed Camera Enforcement:**
 - **Targeted Speed Management:** Expand automated speed enforcement along high-risk corridors and launch targeted safety campaigns aimed at motorcycle speed compliance and helmet usage.

CONCLUSION AND RECOMMENDATIONS

Conclusion

The analysis demonstrates that traffic collisions in New York City represent a systemic public health challenge heavily concentrated by geography, corridor type, and vehicle class. While total crash volume has dropped significantly since 2018, the severity and lethality of individual collisions have escalated substantially. Overwhelmingly driven by preventable human behaviors—primarily driver distraction, tailgating, and failure to yield—collisions pose the highest fatal risk to vulnerable road users like pedestrians and cyclists, who lack physical frame protection against increasingly large passenger SUVs and commercial trucks. Addressing this crisis requires shifting from passive driver messaging to aggressive, forgiving infrastructure redesigns and targeted policy interventions.

Actionable Recommendations for NYC DOT

1. Targeted Capital and Corridor Infrastructure Budgets

- **Reallocate Outer-Borough Funding:** DOT should direct at least 55% to 60% of roadway safety and street redesign capital budgets to high-injury corridors in Brooklyn and Queens (e.g., Atlantic Avenue, Linden Boulevard, Flatbush Avenue Extension).
- **Implement Road Diets on Arterials:** We should re-engineer multi-lane avenues by narrowing vehicle lanes, installing speed cushions, and adding pedestrian refuge islands to reduce open-road speeding.
- **Redesign Gateway & Highway Access Intersections:** Re-align bridge/tunnel approaches and high-volume junctions (such as Flatbush Ave & Tillary St and Bruckner Blvd & E 138th St) with dedicated turn lanes, overhead lane assignment, and flexible dikes to prevent last-second merging cuts.

2. Pedestrian and Micromobility Protection Systems

- **Expand Signal Timing Interventions:** Implement Leading Pedestrian Intervals (LPis) and split-phase turn signals across all top 100 high-incident intersections to eliminate vehicle-pedestrian turning conflicts.
- **Build Protected Micromobility Networks:** Upgrade painted bike lanes to concrete-bollard or curb-protected physical barriers to isolate cyclists, e-bike users, and scooter riders from heavy vehicle traffic.

- **Universal Intersection Daylighting:** Clear parking spaces immediately adjacent to crosswalks across commercial corridors to maximize sightlines between turning drivers and pedestrians.

3. Curbside, Freight, and Vehicle Safety Management

- **Expand Neighborhood Loading Zones (NLZs):** Establish dedicated curbside truck loading zones along high-density commercial corridors (e.g., 2nd and 3rd Avenues) to curb double-parking and unsafe reversing maneuvers.
- **Commercial Direct-Vision & Fleet Standards:** Mandate side guards, blind-spot cameras, and high-visibility cab requirements for city contractor fleets and heavy commercial vehicles operating in dense pedestrian zones.

4. Automated Enforcement and Data Modernization

- **24/7 Automated Camera Coverage:** Expand automated speed safety cameras and red-light enforcement along outer-borough arterial avenues where speeding ranks among top crash causes.
- **Upgrade NYPD Electronic Reporting Software:** Re-engineer police crash logging interfaces to capture precise GPS coordinates automatically and provide distinct sub-categories for distracted driving (e.g., handheld phone vs. texting) to eliminate broad "catch-all" data gaps.